

When you memorize this table, it is helpful to notice that the minus signs go with the derivatives of the “cofunctions,” that is, cosine, cosecant, and cotangent.

Derivatives of Trigonometric Functions

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

EXAMPLE 7 | Differentiate $f(x) = \frac{\sec x}{1 + \tan x}$. For what values of x does the graph of f have a horizontal tangent?

SOLUTION The Quotient Rule gives

$$\begin{aligned} f'(x) &= \frac{(1 + \tan x) \frac{d}{dx}(\sec x) - \sec x \frac{d}{dx}(1 + \tan x)}{(1 + \tan x)^2} \\ &= \frac{(1 + \tan x) \sec x \tan x - \sec x \cdot \sec^2 x}{(1 + \tan x)^2} \\ &= \frac{\sec x (\tan x + \tan^2 x - \sec^2 x)}{(1 + \tan x)^2} \\ &= \frac{\sec x (\tan x - 1)}{(1 + \tan x)^2} \end{aligned}$$

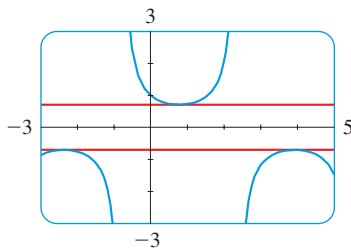


FIGURE 6
The horizontal tangents in Example 7

In simplifying the answer we have used the identity $\tan^2 x + 1 = \sec^2 x$.

Since $\sec x$ is never 0, we see that $f'(x) = 0$ when $\tan x = 1$, and this occurs when $x = n\pi + \pi/4$, where n is an integer (see Figure 6).

EXERCISES 3.4

- Find the derivative of $f(x) = (1 + 2x^2)(x - x^2)$ in two ways: by using the Product Rule and by performing the multiplication first. Do your answers agree?

- Find the derivative of the function

$$F(x) = \frac{x^4 - 5x^3 + \sqrt{x}}{x^2}$$

in two ways: by using the Quotient Rule and by simplifying first. Show that your answers are equivalent. Which method do you prefer?

3–34 Differentiate.

- $f(x) = (x^3 + 2x)e^x$

- $g(x) = \sqrt{x} e^x$

- $g(t) = t^3 \cos t$

- $h(t) = e^t \sin t$

- $F(y) = \left(\frac{1}{y^2} - \frac{3}{y^4}\right)(y + 5y^3)$

- $R(t) = (t + e^t)(3 - \sqrt{t})$

- $f(x) = \sin x + \frac{1}{2} \cot x$

- $h(\theta) = \theta \csc \theta - \cot \theta$

- $y = \frac{e^x}{x^2}$

- $g(x) = \frac{3x - 1}{2x + 1}$

- $y = \frac{x^3}{1 - x^2}$

- $y = \frac{t^2 + 2}{t^4 - 3t^2 + 1}$

- $y = 2 \csc x + 5 \cos x$

- $y = u(a \cos u + b \cot u)$

- $y = \frac{e^x}{1 + x}$

- $f(t) = \frac{2t}{4 + t^2}$

- $y = \frac{x + 1}{x^3 + x - 2}$

- $y = \frac{t}{(t - 1)^2}$

21. $y = (r^2 - 2r)e^r$
22. $y = \frac{1}{s + ke^s}$
23. $f(\theta) = \frac{\sec \theta}{1 + \sec \theta}$
24. $y = \frac{1 + \sin x}{x + \cos x}$
25. $y = \frac{\sin x}{x^2}$
26. $y = \frac{1 - \sec x}{\tan x}$
27. $y = \frac{v^3 - 2v\sqrt{v}}{v}$
28. $z = w^{3/2}(w + ce^w)$
29. $f(t) = \frac{2t}{2 + \sqrt{t}}$
30. $g(t) = \frac{t - \sqrt{t}}{t^{1/3}}$
31. $f(x) = \frac{A}{B + Ce^x}$
32. $f(x) = \frac{1 - xe^x}{x + e^x}$
33. $f(x) = \frac{x}{x + \frac{c}{x}}$
34. $f(x) = \frac{ax + b}{cx + d}$

35–36 Find an equation of the tangent line to the given curve at the specified point.

35. $y = \frac{2x}{x + 1}, (1, 1)$
36. $y = e^x \cos x, (0, 1)$

37–38 Find equations of the tangent line and normal line to the given curve at the specified point.

37. $y = 2xe^x, (0, 0)$
38. $y = \frac{\sqrt{x}}{x + 1}, (4, 0.4)$

39–40 Find $f'(x)$ and $f''(x)$.

39. $f(x) = x^4 e^x$
40. $f(x) = \frac{x}{x^2 - 1}$

41. If $H(\theta) = \theta \sin \theta$, find $H'(\theta)$ and $H''(\theta)$.
42. If $f(t) = \csc t$, find $f''(\pi/6)$.
43. If $f(x) = xe^x$, find the n th derivative, $f^{(n)}(x)$.
44. If $g(x) = x/e^x$, find $g^{(n)}(x)$.

45. Prove that $\frac{d}{dx}(\csc x) = -\csc x \cot x$.

46. Prove that $\frac{d}{dx}(\sec x) = \sec x \tan x$.

47. Prove that $\frac{d}{dx}(\cot x) = -\csc^2 x$.

48. Suppose that $f(\pi/3) = 4$ and $f'(\pi/3) = -2$, and let $g(x) = f(x) \sin x$ and $h(x) = (\cos x)/f(x)$. Find
- (a) $g'(\pi/3)$ (b) $h'(\pi/3)$

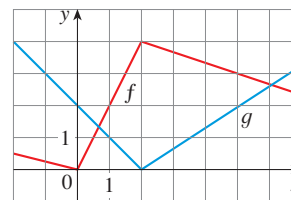
49. Suppose that $f(5) = 1$, $f'(5) = 6$, $g(5) = -3$, and $g'(5) = 2$. Find the following values.
- (a) $(fg)'(5)$ (b) $(f/g)'(5)$ (c) $(g/f)'(5)$

50. Suppose that $f(2) = -3$, $g(2) = 4$, $f'(2) = -2$, and $g'(2) = 7$. Find $h'(2)$.

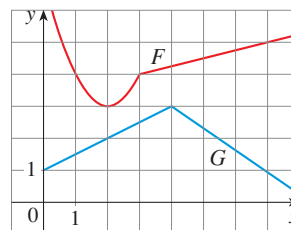
- (a) $h(x) = 5f(x) - 4g(x)$ (b) $h(x) = f(x)g(x)$
- (c) $h(x) = \frac{f(x)}{g(x)}$ (d) $h(x) = \frac{g(x)}{1 + f(x)}$

51. If f and g are the functions whose graphs are shown, let $u(x) = f(x)g(x)$ and $v(x) = f(x)/g(x)$.

- (a) Find $u'(1)$. (b) Find $v'(5)$.



52. Let $P(x) = F(x)G(x)$ and $Q(x) = F(x)/G(x)$, where F and G are the functions whose graphs are shown.
- (a) Find $P'(2)$. (b) Find $Q'(7)$.



53. If g is a differentiable function, find an expression for the derivative of each of the following functions.

- (a) $y = xg(x)$ (b) $y = \frac{x}{g(x)}$ (c) $y = \frac{g(x)}{x}$

54. **Insecticide resistance** If the frequency of a gene for insecticide resistance is p , then its frequency in the next generation is given by the expression

$$\frac{p(1 + s)}{1 + sp}$$

where s is the reproductive advantage that this gene has over the wild type in the presence of the insecticide. Determine the rate at which the gene frequency in the next generation changes as s changes.

55. The **Michaelis-Menten equation** for the enzyme chymotrypsin is

$$v = \frac{0.14[S]}{0.015 + [S]}$$

where v is the rate of an enzymatic reaction and $[S]$ is the concentration of a substrate S . Calculate $dv/d[S]$ and interpret it.

56. The **biomass** $B(t)$ of a fish population is the total mass of the members of the population at time t . It is the product of the number of individuals $N(t)$ in the population and the average mass $M(t)$ of a fish at time t . In the case of guppies, breeding occurs continually. Suppose that at time $t = 4$ weeks the population is 820 guppies and is growing at a rate of 50 guppies per week, while the average mass is 1.2 g and is increasing at a rate of 0.14 g/week. At what rate is the biomass increasing when $t = 4$?
57. The gas law for an ideal gas at absolute temperature T (in kelvins), pressure P (in atmospheres), and volume V (in liters) is $PV = nRT$, where n is the number of moles of the gas and $R = 0.0821$ is the gas constant. Suppose that, at a certain instant, $P = 8.0$ atm and is increasing at a rate of 0.10 atm/min and $V = 10$ L and is decreasing at a rate of 0.15 L/min. Find the rate of change of T with respect to time at that instant if $n = 10$ mol.
58. **Sensitivity of the eye to brightness** If R denotes the reaction of the body to some stimulus of strength x , the *sensitivity* S is defined to be the rate of change of the reaction with respect to x . A particular example is that when the brightness x of a light source is increased, the eye reacts by decreasing the area R of the pupil. The experimental formula

$$R = \frac{40 + 24x^{0.4}}{1 + 4x^{0.4}}$$

has been used to model the dependence of R on x when R is measured in square millimeters and x is measured in appropriate units of brightness.

(a) Find the sensitivity.



(b) Illustrate part (a) by graphing both R and S as functions of x . Comment on the values of R and S at low levels of brightness. Is this what you would expect?

59. How many tangent lines to the curve $y = x/(x + 1)$ pass through the point $(1, 2)$? At which points do these tangent lines touch the curve?

60. Find equations of the tangent lines to the curve

$$y = \frac{x - 1}{x + 1}$$

that are parallel to the line $x - 2y = 2$.

61. (a) Use the Product Rule twice to prove that if f , g , and h are differentiable, then $(fgh)' = f'gh + fg'h + fgh'$.
(b) Taking $f = g = h$ in part (a), show that

$$\frac{d}{dx} [f(x)]^3 = 3[f(x)]^2 f'(x)$$

(c) Use part (b) to differentiate $y = e^{3x}$.

62. (a) If $F(x) = f(x)g(x)$, where f and g have derivatives of all orders, show that $F'' = f''g + 2f'g' + fg''$.
(b) Find similar formulas for F''' and $F^{(4)}$.
(c) Guess a formula for $F^{(n)}$.

63. (a) If g is differentiable, the **Reciprocal Rule** says that

$$\frac{d}{dx} \left[\frac{1}{g(x)} \right] = -\frac{g'(x)}{[g(x)]^2}$$

Use the Quotient Rule to prove the Reciprocal Rule.

- (b) Use the Reciprocal Rule to differentiate the function $y = 1/(x^4 + x^2 + 1)$.
(c) Use the Reciprocal Rule to verify that the Power Rule is valid for negative integers, that is,

$$\frac{d}{dx} (x^{-n}) = -nx^{-n-1}$$

for all positive integers n .

3.5 The Chain Rule

Suppose you are asked to differentiate the function

$$F(x) = \sqrt{x^2 + 1}$$

The differentiation formulas you learned in the previous sections of this chapter do not enable you to calculate $F'(x)$.

Observe that F is a composite function. In fact, if we let $y = f(u) = \sqrt{u}$ and let $u = g(x) = x^2 + 1$, then we can write $y = F(x) = f(g(x))$, that is, $F = f \circ g$. We know how to differentiate both f and g , so it would be useful to have a rule that tells us how to find the derivative of $F = f \circ g$ in terms of the derivatives of f and g .

It turns out that the derivative of the composite function $f \circ g$ is the product of the derivatives of f and g . This fact is one of the most important of the differentiation rules and is called the *Chain Rule*. It seems plausible if we interpret derivatives as rates of change. Regard du/dx as the rate of change of u with respect to x , dy/du as the rate of change of y with respect to u , and dy/dx as the rate of change of y with respect to x . If u changes

See Section 1.3 for a review of composite functions.

twice as fast as x and y changes three times as fast as u , then it seems reasonable that y changes six times as fast as x , and so we expect that

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

The Chain Rule If g is differentiable at x and f is differentiable at $g(x)$, then the composite function $F = f \circ g$ defined by $F(x) = f(g(x))$ is differentiable at x and F' is given by the product

$$F'(x) = f'(g(x)) \cdot g'(x)$$

In Leibniz notation, if $y = f(u)$ and $u = g(x)$ are both differentiable functions, then

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

COMMENTS ON THE PROOF OF THE CHAIN RULE Let Δu be the change in u corresponding to a change of Δx in x , that is,

$$\Delta u = g(x + \Delta x) - g(x)$$

Then the corresponding change in y is

$$\Delta y = f(u + \Delta u) - f(u)$$

It is tempting to write

$$\begin{aligned} \frac{dy}{dx} &= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} \\ (1) \quad &= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta u} \cdot \frac{\Delta u}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta u} \cdot \lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} \\ &= \lim_{\Delta u \rightarrow 0} \frac{\Delta y}{\Delta u} \cdot \lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} \quad (\text{Note that } \Delta u \rightarrow 0 \text{ as } \Delta x \rightarrow 0 \text{ since } g \text{ is continuous.}) \\ &= \frac{dy}{du} \frac{du}{dx} \end{aligned}$$

The only flaw in this reasoning is that in (1) it might happen that $\Delta u = 0$ (even when $\Delta x \neq 0$) and, of course, we can't divide by 0. Nonetheless, this reasoning does at least suggest that the Chain Rule is true. A full proof of the Chain Rule is given at the end of this section. ■

The Chain Rule can be written either in the prime notation

$$(2) \quad (f \circ g)'(x) = f'(g(x)) \cdot g'(x)$$

or, if $y = f(u)$ and $u = g(x)$, in Leibniz notation:

$$(3) \quad \frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

James Gregory

The first person to formulate the Chain Rule was the Scottish mathematician James Gregory (1638–1675), who also designed the first practical reflecting telescope. Gregory discovered the basic ideas of calculus at about the same time as Newton. He became the first Professor of Mathematics at the University of St. Andrews and later held the same position at the University of Edinburgh. But one year after accepting that position he died at the age of 36.

Equation 3 is easy to remember because if dy/du and du/dx were quotients, then we could cancel du . Remember, however, that du has not been defined and du/dx should not be thought of as an actual quotient.

EXAMPLE 1 | Find $F'(x)$ if $F(x) = \sqrt{x^2 + 1}$.

SOLUTION 1 (using Equation 2): At the beginning of this section we expressed F as $F(x) = (f \circ g)(x) = f(g(x))$ where $f(u) = \sqrt{u}$ and $g(x) = x^2 + 1$. Since

$$f'(u) = \frac{1}{2}u^{-1/2} = \frac{1}{2\sqrt{u}} \quad \text{and} \quad g'(x) = 2x$$

we have

$$\begin{aligned} F'(x) &= f'(g(x)) \cdot g'(x) \\ &= \frac{1}{2\sqrt{x^2 + 1}} \cdot 2x = \frac{x}{\sqrt{x^2 + 1}} \end{aligned}$$

SOLUTION 2 (using Equation 3): If we let $u = x^2 + 1$ and $y = \sqrt{u}$, then

$$F'(x) = \frac{dy}{du} \frac{du}{dx} = \frac{1}{2\sqrt{u}} (2x) = \frac{1}{2\sqrt{x^2 + 1}} (2x) = \frac{x}{\sqrt{x^2 + 1}} \quad \blacksquare$$

When using Formula 3 we should bear in mind that dy/dx refers to the derivative of y when y is considered as a function of x (called the *derivative of y with respect to x*), whereas dy/du refers to the derivative of y when considered as a function of u (the derivative of y with respect to u). For instance, in Example 1, y can be considered as a function of x ($y = \sqrt{x^2 + 1}$) and also as a function of u ($y = \sqrt{u}$). Note that

$$\frac{dy}{dx} = F'(x) = \frac{x}{\sqrt{x^2 + 1}} \quad \text{whereas} \quad \frac{dy}{du} = f'(u) = \frac{1}{2\sqrt{u}}$$

NOTE In using the Chain Rule we work from the outside to the inside. Formula 2 says that *we differentiate the outer function f [at the inner function $g(x)$] and then we multiply by the derivative of the inner function.*

$$\frac{d}{dx} \underbrace{f}_{\text{outer function}} \underbrace{(g(x))}_{\text{evaluated at inner function}} = \underbrace{f'}_{\text{derivative of outer function}} \underbrace{(g(x))}_{\text{evaluated at inner function}} \cdot \underbrace{g'(x)}_{\text{derivative of inner function}}$$

EXAMPLE 2 | Differentiate (a) $y = \sin(x^2)$ and (b) $y = \sin^2 x$.

SOLUTION

(a) If $y = \sin(x^2)$, then the outer function is the sine function and the inner function is the squaring function, so the Chain Rule gives

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} \underbrace{\sin}_{\text{outer function}} \underbrace{(x^2)}_{\text{evaluated at inner function}} = \underbrace{\cos}_{\text{derivative of outer function}} \underbrace{(x^2)}_{\text{evaluated at inner function}} \cdot \underbrace{2x}_{\text{derivative of inner function}} \\ &= 2x \cos(x^2) \end{aligned}$$

(b) Note that $\sin^2 x = (\sin x)^2$. Here the outer function is the squaring function and the inner function is the sine function. So

$$\frac{dy}{dx} = \frac{d}{dx} (\underbrace{\sin x}_{\text{inner function}})^2 = \underbrace{2}_{\text{derivative of outer function}} \cdot \underbrace{(\sin x)}_{\text{evaluated at inner function}} \cdot \underbrace{\cos x}_{\text{derivative of inner function}}$$

See Reference Page 2 or Appendix C.

The answer can be left as $2 \sin x \cos x$ or written as $\sin 2x$ (by a trigonometric identity known as the double-angle formula). ■

■ Combining the Chain Rule with Other Rules

In Example 2(a) we combined the Chain Rule with the rule for differentiating the sine function. In general, if $y = \sin u$, where u is a differentiable function of x , then, by the Chain Rule,

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = \cos u \frac{du}{dx}$$

Thus
$$\frac{d}{dx} (\sin u) = \cos u \frac{du}{dx}$$

In a similar fashion, all of the differentiation formulas for trigonometric functions can be combined with the Chain Rule.

Let's make explicit the special case of the Chain Rule where the outer function f is a power function. If $y = [g(x)]^n$, then we can write $y = f(u) = u^n$ where $u = g(x)$. By using the Chain Rule and then the Power Rule, we get

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = nu^{n-1} \frac{du}{dx} = n[g(x)]^{n-1} g'(x)$$

(4) The Power Rule Combined with the Chain Rule If n is any real number and $u = g(x)$ is differentiable, then

$$\frac{d}{dx} (u^n) = nu^{n-1} \frac{du}{dx}$$

Alternatively,
$$\frac{d}{dx} [g(x)]^n = n[g(x)]^{n-1} \cdot g'(x)$$

Notice that the derivative in Example 1 could be calculated by taking $n = \frac{1}{2}$ in Rule 4.

EXAMPLE 3 | Differentiate $y = (x^3 - 1)^{100}$.

SOLUTION Taking $u = g(x) = x^3 - 1$ and $n = 100$ in (4), we have

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} (x^3 - 1)^{100} = 100(x^3 - 1)^{99} \frac{d}{dx} (x^3 - 1) \\ &= 100(x^3 - 1)^{99} \cdot 3x^2 = 300x^2(x^3 - 1)^{99} \end{aligned}$$

EXAMPLE 4 | Find $f'(x)$ if $f(x) = \frac{1}{\sqrt[3]{x^2 + x + 1}}$.

SOLUTION First rewrite f : $f(x) = (x^2 + x + 1)^{-1/3}$

$$\begin{aligned}\text{Thus } f'(x) &= -\frac{1}{3}(x^2 + x + 1)^{-4/3} \frac{d}{dx}(x^2 + x + 1) \\ &= -\frac{1}{3}(x^2 + x + 1)^{-4/3}(2x + 1)\end{aligned}$$

EXAMPLE 5 | Find the derivative of the function

$$g(t) = \left(\frac{t-2}{2t+1} \right)^9$$

SOLUTION Combining the Power Rule, Chain Rule, and Quotient Rule, we get

$$\begin{aligned}g'(t) &= 9 \left(\frac{t-2}{2t+1} \right)^8 \frac{d}{dt} \left(\frac{t-2}{2t+1} \right) \\ &= 9 \left(\frac{t-2}{2t+1} \right)^8 \frac{(2t+1) \cdot 1 - 2(t-2)}{(2t+1)^2} = \frac{45(t-2)^8}{(2t+1)^{10}}\end{aligned}$$

EXAMPLE 6 | Differentiate $y = (2x + 1)^5(x^3 - x + 1)^4$.

SOLUTION In this example we must use the Product Rule before using the Chain Rule:

$$\begin{aligned}\frac{dy}{dx} &= (2x + 1)^5 \frac{d}{dx}(x^3 - x + 1)^4 + (x^3 - x + 1)^4 \frac{d}{dx}(2x + 1)^5 \\ &= (2x + 1)^5 \cdot 4(x^3 - x + 1)^3 \frac{d}{dx}(x^3 - x + 1) \\ &\quad + (x^3 - x + 1)^4 \cdot 5(2x + 1)^4 \frac{d}{dx}(2x + 1) \\ &= 4(2x + 1)^5(x^3 - x + 1)^3(3x^2 - 1) + 5(x^3 - x + 1)^4(2x + 1)^4 \cdot 2\end{aligned}$$

Noticing that each term has the common factor $2(2x + 1)^4(x^3 - x + 1)^3$, we could factor it out and write the answer as

$$\frac{dy}{dx} = 2(2x + 1)^4(x^3 - x + 1)^3(17x^3 + 6x^2 - 9x + 3)$$

EXAMPLE 7 | Differentiate $y = e^{\sin x}$.

SOLUTION Here the inner function is $g(x) = \sin x$ and the outer function is the exponential function $f(x) = e^x$. So, by the Chain Rule,

$$\frac{dy}{dx} = \frac{d}{dx}(e^{\sin x}) = e^{\sin x} \frac{d}{dx}(\sin x) = e^{\sin x} \cos x$$

The graphs of the functions y and y' in Example 6 are shown in Figure 1. Notice that y' is large when y increases rapidly and $y' = 0$ when y has a horizontal tangent. So our answer appears to be reasonable.

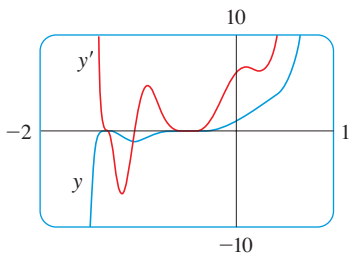
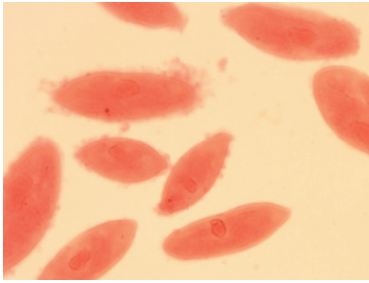


FIGURE 1



EXAMPLE 8 | **BB** **Gause's logistic model** In an experiment with the protozoan *Paramecium*, the biologist G. F. Gause modeled the protozoan population size with the logistic function

$$P(t) = \frac{61}{1 + 31e^{-0.7944t}}$$

where t is measured in days. According to this model, how fast was the population growing after 8 days?

SOLUTION We use the Reciprocal Rule as in Exercise 3.4.63 (or the Quotient Rule) and the Chain Rule to differentiate P :

$$\begin{aligned} P'(t) &= -\frac{61}{(1 + 31e^{-0.7944t})^2} \cdot \frac{d}{dt}(1 + 31e^{-0.7944t}) \\ &= -\frac{61 \cdot 31 \cdot (-0.7944)e^{-0.7944t}}{(1 + 31e^{-0.7944t})^2} \approx \frac{1502e^{-0.7944t}}{(1 + 31e^{-0.7944t})^2} \end{aligned}$$

When $t = 8$, we have

$$P'(8) = \frac{1502e^{-8(0.7944)}}{(1 + 31e^{-8(0.7944)})^2} \approx 2.35$$

So at that time the population was increasing at a rate of about two per day. ■

■ Exponential Functions with Arbitrary Bases

We can use the Chain Rule to differentiate an exponential function with any base $b > 0$. Recall from Section 1.5 that $b = e^{\ln b}$. So

$$b^x = (e^{\ln b})^x = e^{(\ln b)x}$$

and the Chain Rule gives

$$\begin{aligned} \frac{d}{dx}(b^x) &= \frac{d}{dx}(e^{(\ln b)x}) = e^{(\ln b)x} \frac{d}{dx}(\ln b)x \\ &= e^{(\ln b)x} \cdot \ln b = b^x \ln b \end{aligned}$$

because $\ln b$ is a constant. So we have the formula

Don't confuse Formula 5 (where x is the exponent) with the Power Rule (where x is the base):

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

(5)

$$\frac{d}{dx}(b^x) = b^x \ln b$$

In particular, if $b = 2$, we get

(6)

$$\frac{d}{dx}(2^x) = 2^x \ln 2$$

In Section 3.3 we gave the estimate

$$\frac{d}{dx}(2^x) \approx (0.69)2^x$$

This is consistent with the exact formula (6) because $\ln 2 \approx 0.693147$.

EXAMPLE 9 | **Viral load** In Section 1.4 we considered the function

$$V(t) = 96.39785 \cdot (0.818656)^t$$

for the viral load in a patient with HIV after treatment with ABT-538. Find $V'(10)$ and interpret it.

SOLUTION Using Formula 5 with $b = 0.818656$, we get

$$\begin{aligned} V'(t) &= 96.39785 \cdot (0.818656)^t \ln(0.818656) \\ &\approx -19.288 \cdot (0.818656)^t \end{aligned}$$

So
$$V'(10) \approx -19.288 \cdot (0.818656)^{10} \approx -2.6$$

This means that after 10 days the viral load was decreasing at a rate of about 2.6 RNA copies/mL per day. ■

■ Longer Chains

The reason for the name “Chain Rule” becomes clear when we make a longer chain by adding another link. Suppose that $y = f(u)$, $u = g(x)$, and $x = h(t)$, where f , g , and h are differentiable functions. Then, to compute the derivative of y with respect to t , we use the Chain Rule twice:

$$\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt} = \frac{dy}{du} \frac{du}{dx} \frac{dx}{dt}$$

EXAMPLE 10 | If $f(x) = \sin(\cos(\tan x))$, then

$$\begin{aligned} f'(x) &= \cos(\cos(\tan x)) \frac{d}{dx} \cos(\tan x) \\ &= \cos(\cos(\tan x)) [-\sin(\tan x)] \frac{d}{dx} (\tan x) \\ &= -\cos(\cos(\tan x)) \sin(\tan x) \sec^2 x \end{aligned}$$

Notice that we used the Chain Rule twice. ■

EXAMPLE 11 | Differentiate $y = e^{\sec 3\theta}$.

SOLUTION The outer function is the exponential function, the middle function is the secant function, and the inner function is the tripling function. So we have

$$\begin{aligned} \frac{dy}{d\theta} &= e^{\sec 3\theta} \frac{d}{d\theta} (\sec 3\theta) \\ &= e^{\sec 3\theta} \sec 3\theta \tan 3\theta \frac{d}{d\theta} (3\theta) \\ &= 3e^{\sec 3\theta} \sec 3\theta \tan 3\theta \end{aligned}$$

■

■ Implicit Differentiation

The functions that we have met so far can be described by expressing one variable explicitly in terms of another variable—for example,

$$y = \sqrt{x^3 + 1} \quad \text{or} \quad y = x \sin x$$

or, in general, $y = f(x)$. Some functions, however, are defined implicitly by a relation between x and y such as

$$(7) \quad x^3 + y^3 = 6xy$$

It's not easy to solve Equation 7 for y explicitly as a function of x by hand. (A computer algebra system has no trouble, but the expressions it obtains are very complicated.) Nonetheless, (7) is the equation of a curve called the **folium of Descartes** shown in Figure 2 and it implicitly defines y as several functions of x . The graphs of three such functions are shown in Figure 3. When we say that f is a function defined implicitly by Equation 2, we mean that the equation

$$x^3 + [f(x)]^3 = 6xf(x)$$

is true for all values of x in the domain of f .

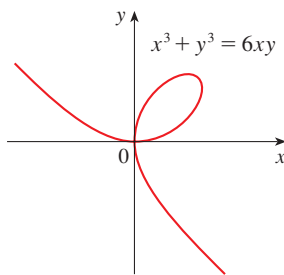


FIGURE 2
The folium of Descartes

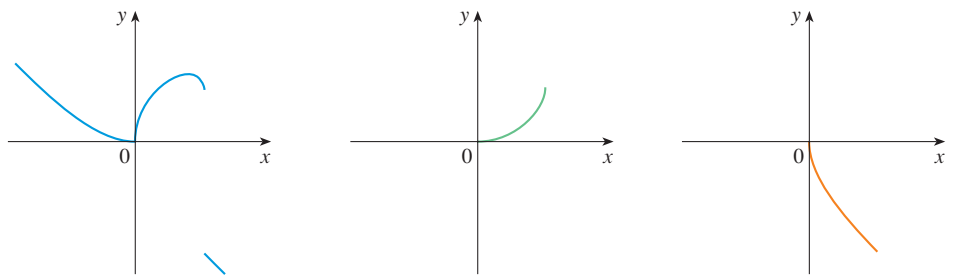


FIGURE 3
Graphs of three functions defined by the folium of Descartes

Fortunately, we don't need to solve an equation for y in terms of x in order to find the derivative of y . Instead we can use the method of **implicit differentiation**. This consists of differentiating both sides of the equation with respect to x and then solving the resulting equation for y' .

EXAMPLE 12

- (a) Find y' if $x^3 + y^3 = 6xy$.
 (b) Find the tangent to the folium of Descartes $x^3 + y^3 = 6xy$ at the point $(3, 3)$.

SOLUTION

- (a) Differentiating both sides of $x^3 + y^3 = 6xy$ with respect to x , we have

$$\frac{d}{dx}(x^3 + y^3) = \frac{d}{dx}(6xy)$$

Remembering that y is a function of x , and using the Chain Rule on the term y^3 and the Product Rule on the term $6xy$, we get

$$3x^2 + 3y^2y' = 6xy' + 6y$$

or

$$x^2 + y^2y' = 2xy' + 2y$$

We now solve for y' :

$$y^2y' - 2xy' = 2y - x^2$$

$$(y^2 - 2x)y' = 2y - x^2$$

$$y' = \frac{2y - x^2}{y^2 - 2x}$$

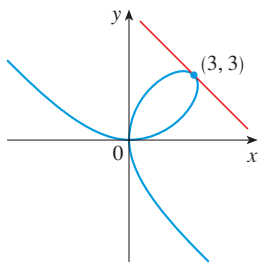


FIGURE 4

(b) When $x = y = 3$,

$$y' = \frac{2 \cdot 3 - 3^2}{3^2 - 2 \cdot 3} = -1$$

and a glance at Figure 4 confirms that this is a reasonable value for the slope at $(3, 3)$. So an equation of the tangent to the folium at $(3, 3)$ is

$$y - 3 = -1(x - 3) \quad \text{or} \quad x + y = 6$$

EXAMPLE 13 | **BB** Infectious disease outbreak size Mathematical models have been used to predict the fraction of a population that will be infected by a disease under different conditions. The Kermack-McKendrick model (see Exercise 7.6.23) leads to the following equation:

$$\rho e^{-qA} = 1 - A$$

where A is the fraction of the population ultimately infected, q is a measure of disease transmissibility, and ρ is a measure of the fraction of the population that is initially susceptible to infection. How does the outbreak size A change with an increase in transmissibility q ?

SOLUTION The given equation can't be solved for A , so we use implicit differentiation to find dA/dq :

$$\frac{d}{dq} (\rho e^{-qA}) = \frac{d}{dq} (1 - A)$$

Keeping in mind that A is a function of q , we have

$$\rho e^{-qA} \cdot \frac{d}{dq} (-qA) = -\frac{dA}{dq}$$

Now we use the Product Rule:

$$\rho e^{-qA} \left(-q \frac{dA}{dq} - A \cdot 1 \right) = -\frac{dA}{dq}$$

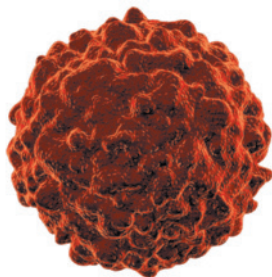
Solving for dA/dq , we get

$$\begin{aligned} (-\rho q e^{-qA} + 1) \frac{dA}{dq} &= A \rho e^{-qA} \\ \frac{dA}{dq} &= \frac{A \rho e^{-qA}}{1 - \rho q e^{-qA}} = \frac{A \rho}{e^{qA} - \rho q} \end{aligned}$$

This gives the rate of increase of the outbreak size as the transmissibility changes. ■

■ Related Rates

In a related rates problem the idea is to compute the rate of change of one quantity in terms of the rate of change of another quantity. The procedure is to find an equation that relates the two quantities and then use the Chain Rule to differentiate both sides with respect to time.



EXAMPLE 14 | **BB** **Growth of a tumor** When the diameter of a spherical tumor is 16 mm it is growing at a rate of 0.4 mm a day. How fast is the volume of the tumor changing at that time?

SOLUTION We model the shape of the tumor by a sphere. If r is the radius of the sphere, then its volume is

$$V = \frac{4}{3}\pi r^3$$

In order to use the given information, we differentiate each side of this equation with respect to t (time). To differentiate the right side we need to use the Chain Rule:

$$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt} = 4\pi r^2 \frac{dr}{dt}$$

At the time in question, the radius is $r = \frac{1}{2}(16) = 8$ mm and it is increasing at a rate of

$$\frac{dr}{dt} = \frac{1}{2}(0.4) = 0.2 \text{ mm/day}$$

So the rate of increase of the volume of the tumor is

$$\frac{dV}{dt} = 4\pi(8)^2 \cdot 0.2 = 51.2\pi \approx 161 \text{ mm}^3/\text{day}$$

■ How To Prove the Chain Rule

Recall that if $y = f(x)$ and x changes from a to $a + \Delta x$, we define the increment of y as

$$\Delta y = f(a + \Delta x) - f(a)$$

According to the definition of a derivative, we have

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = f'(a)$$

So if we denote by ε the difference between the difference quotient and the derivative, we obtain

$$\lim_{\Delta x \rightarrow 0} \varepsilon = \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta y}{\Delta x} - f'(a) \right) = f'(a) - f'(a) = 0$$

But

$$\varepsilon = \frac{\Delta y}{\Delta x} - f'(a) \quad \Rightarrow \quad \Delta y = f'(a) \Delta x + \varepsilon \Delta x$$

If we define ε to be 0 when $\Delta x = 0$, then ε becomes a continuous function of Δx . Thus, for a differentiable function f , we can write

$$(8) \quad \Delta y = f'(a) \Delta x + \varepsilon \Delta x \quad \text{where} \quad \varepsilon \rightarrow 0 \text{ as } \Delta x \rightarrow 0$$

and ε is a continuous function of Δx . This property of differentiable functions is what enables us to prove the Chain Rule.

PROOF OF THE CHAIN RULE Suppose $u = g(x)$ is differentiable at a and $y = f(u)$ is differentiable at $b = g(a)$. If Δx is an increment in x and Δu and Δy are the corresponding increments in u and y , then we can use Equation 8 to write

$$(9) \quad \Delta u = g'(a) \Delta x + \varepsilon_1 \Delta x = [g'(a) + \varepsilon_1] \Delta x$$

where $\varepsilon_1 \rightarrow 0$ as $\Delta x \rightarrow 0$. Similarly

$$(10) \quad \Delta y = f'(b) \Delta u + \varepsilon_2 \Delta u = [f'(b) + \varepsilon_2] \Delta u$$

where $\varepsilon_2 \rightarrow 0$ as $\Delta u \rightarrow 0$. If we now substitute the expression for Δu from Equation 9 into Equation 10, we get

$$\Delta y = [f'(b) + \varepsilon_2][g'(a) + \varepsilon_1] \Delta x$$

$$\text{so} \quad \frac{\Delta y}{\Delta x} = [f'(b) + \varepsilon_2][g'(a) + \varepsilon_1]$$

As $\Delta x \rightarrow 0$, Equation 9 shows that $\Delta u \rightarrow 0$. So both $\varepsilon_1 \rightarrow 0$ and $\varepsilon_2 \rightarrow 0$ as $\Delta x \rightarrow 0$. Therefore

$$\begin{aligned} \frac{dy}{dx} &= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} [f'(b) + \varepsilon_2][g'(a) + \varepsilon_1] \\ &= f'(b)g'(a) = f'(g(a))g'(a) \end{aligned}$$

This proves the Chain Rule. ■

EXERCISES 3.5

1–6 Write the composite function in the form $f(g(x))$. [Identify the inner function $u = g(x)$ and the outer function $y = f(u)$.] Then find the derivative dy/dx .

1. $y = \sqrt[3]{1 + 4x}$
2. $y = (2x^3 + 5)^4$
3. $y = \tan \pi x$
4. $y = \sin(\cot x)$
5. $y = e^{\sqrt{x}}$
6. $y = \sqrt{2 - e^x}$

7–36 Find the derivative of the function.

7. $F(x) = (x^4 + 3x^2 - 2)^5$
8. $F(x) = (4x - x^2)^{100}$
9. $F(x) = \sqrt{1 - 2x}$
10. $f(x) = (1 + x^4)^{2/3}$
11. $f(z) = \frac{1}{z^2 + 1}$
12. $f(t) = \sqrt[3]{1 + \tan t}$
13. $y = \cos(a^3 + x^3)$
14. $y = a^3 + \cos^3 x$
15. $h(t) = t^3 - 3^t$
16. $y = 3 \cot(n\theta)$
17. $y = xe^{-kx}$
18. $y = e^{-2t} \cos 4t$
19. $y = (2x - 5)^4(8x^2 - 5)^{-3}$
20. $h(t) = (t^4 - 1)^3(t^3 + 1)^4$
21. $y = e^{x \cos x}$
22. $y = 10^{1-x^2}$
23. $y = \left(\frac{x^2 + 1}{x^2 - 1} \right)^3$
24. $G(y) = \left(\frac{y^2}{y + 1} \right)^5$
25. $y = \sec^2 x + \tan^2 x$
26. $y = \frac{e^u - e^{-u}}{e^u + e^{-u}}$

$$27. y = \frac{r}{\sqrt{r^2 + 1}}$$

$$29. y = \sin(\tan 2x)$$

$$31. y = 2^{\sin \pi x}$$

$$33. y = \cot^2(\sin \theta)$$

$$35. y = \cos \sqrt{\sin(\tan \pi x)}$$

$$28. y = e^{k \tan \sqrt{x}}$$

$$30. f(t) = \sqrt{\frac{t}{t^2 + 4}}$$

$$32. y = \sin(\sin(\sin x))$$

$$34. y = \sqrt{x + \sqrt{x + \sqrt{x}}}$$

$$36. y = 2^{3x^2}$$

37–40 Find y' and y'' .

$$37. y = \cos(x^2)$$

$$39. y = e^{\alpha x} \sin \beta x$$

$$38. y = \cos^2 x$$

$$40. y = e^{e^x}$$

41–44 Find an equation of the tangent line to the curve at the given point.

$$41. y = (1 + 2x)^{10}, \quad (0, 1)$$

$$43. y = \sin(\sin x), \quad (\pi, 0)$$

$$42. y = \sqrt{1 + x^3}, \quad (2, 3)$$

$$44. y = \sin x + \sin^2 x, \quad (0, 0)$$

45. If $F(x) = f(g(x))$, where $f(-2) = 8$, $f'(-2) = 4$, $f'(5) = 3$, $g(5) = -2$, and $g'(5) = 6$, find $F'(5)$.

46. If $h(x) = \sqrt{4 + 3f(x)}$, where $f(1) = 7$ and $f'(1) = 4$, find $h'(1)$.

47. A table of values for f , g , f' , and g' is given.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	3	2	4	6
2	1	8	5	7
3	7	2	7	9

- (a) If $h(x) = f(g(x))$, find $h'(1)$.
 (b) If $H(x) = g(f(x))$, find $H'(1)$.
48. Let f and g be the functions in Exercise 47.
 (a) If $F(x) = f(f(x))$, find $F'(2)$.
 (b) If $G(x) = g(g(x))$, find $G'(3)$.
49. Suppose f is differentiable on $\mathbb{R}$. Let $F(x) = f(e^x)$ and $G(x) = e^{f(x)}$. Find expressions for (a) $F'(x)$ and (b) $G'(x)$.
50. Suppose f is differentiable on $\mathbb{R}$ and α is a real number. Let $F(x) = f(x^\alpha)$ and $G(x) = [f(x)]^\alpha$. Find expressions for (a) $F'(x)$ and (b) $G'(x)$.
51. Let $r(x) = f(g(h(x)))$, where $h(1) = 2$, $g(2) = 3$, $h'(1) = 4$, $g'(2) = 5$, and $f'(3) = 6$. Find $r'(1)$.
52. If g is a twice differentiable function and $f(x) = xg(x^2)$, find f'' in terms of g , g' , and g'' .
53. Find the 50th derivative of $y = \cos 2x$.
54. Find the 1000th derivative of $f(x) = xe^{-x}$.
55. The displacement of a particle on a vibrating string is given by the equation

$$s(t) = 10 + \frac{1}{4} \sin(10\pi t)$$

where s is measured in centimeters and t in seconds. Find the velocity and acceleration of the particle after t seconds.

56. **Oral antibiotics** In Example 1.3.7 we studied a model for antibiotic use in sinus infections. If x is the amount of antibiotic taken orally (in mg), then the function $h(x)$ gives the amount entering the bloodstream through the stomach. If x mg reaches the bloodstream, then $g(x)$ gives the amount that survives filtration by the liver. Finally, if x mg survives filtration by the liver, then $f(x)$ is absorbed into the sinus cavity. Thus, for a given dose x , the amount making it to the sinus cavity is $A(x) = f(g(h(x)))$. Suppose that a dose of 500 mg is given, $h(500) = 8$, $g(8) = 2$, $f(2) = 1.5$, and $h'(500) = 2.5$, $g'(8) = \frac{1}{4}$, and $f'(2) = 1$. Calculate $A'(x)$ and interpret your answer.
57. **Gene regulation** Genes produce molecules called mRNA that go on to produce proteins. High concentrations of protein inhibit the production of mRNA, leading to stable gene regulation. This process has been modeled (see Section 10.3) to show that the concentration of mRNA over time is given by the equation

$$m(t) = \frac{1}{2}e^{-t}(\sin t - \cos t) + \frac{1}{2}$$

What is the rate of change of mRNA concentration as a function of time?

58. **World population growth** In Example 1.4.3 we modeled the world population from 1900 to 2010 with the exponential function

$$P(t) = (1436.53) \cdot (1.01395)^t$$

where $t = 0$ corresponds to the year 1900 and $P(t)$ is measured in millions. According to this model, what was the rate of increase of world population in 1920? In 1950? In 2000?

59. **Blood alcohol concentration** In Section 3.1 we discussed an experiment in which the average BAC of eight male subjects was measured after consumption of 15 mL of ethanol (corresponding to one alcoholic drink). The resulting data were modeled by the concentration function

$$C(t) = 0.0225te^{-0.0467t}$$

where t is measured in minutes after consumption and C is measured in mg/mL.

- (a) How rapidly was the BAC increasing after 10 minutes?
 (b) How rapidly was it decreasing half an hour later?

60. **Logistic growth in Japan** The midyear population in Japan from 1960 to 2010 has been modeled by the function

$$P(t) = 94,000 + \frac{32,658.5}{1 + 12.75e^{-0.1706t}}$$

where t is measured in years since 1960 and $P(t)$ is measured in thousands. According to this model, how quickly was the Japanese population growing in 1970? In 1990?

61. Under certain circumstances a rumor spreads according to the equation

$$p(t) = \frac{1}{1 + ae^{-kt}}$$

where $p(t)$ is the proportion of the population that has heard the rumor at time t and a and k are positive constants.

- (a) Find $\lim_{t \rightarrow \infty} p(t)$.
 (b) Find the rate of spread of the rumor.
 (c) Graph p for the case $a = 10$, $k = 0.5$ with t measured in hours. Use the graph to estimate how long it will take for 80% of the population to hear the rumor.

62. In Example 1.3.4 we arrived at a model for the length of daylight (in hours) in Philadelphia on the t th day of the year:

$$L(t) = 12 + 2.8 \sin \left[\frac{2\pi}{365}(t - 80) \right]$$

Use this model to compare how the number of hours of daylight is increasing in Philadelphia on March 21 and May 21.

63–64

- (a) Find y' by implicit differentiation.
 (b) Solve the equation explicitly for y and differentiate to get y' in terms of x .
 (c) Check that your solutions to parts (a) and (b) are consistent by substituting the expression for y into your solution for part (a).

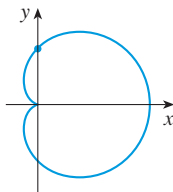
63. $xy + 2x + 3x^2 = 4$ 64. $\cos x + \sqrt{y} = 5$

65–76 Find dy/dx by implicit differentiation.

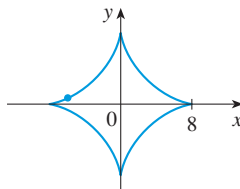
65. $x^3 + y^3 = 1$ 66. $2\sqrt{x} + \sqrt{y} = 3$
 67. $x^2 + xy - y^2 = 4$ 68. $2x^3 + x^2y - xy^3 = 2$
 69. $x^4(x + y) = y^2(3x - y)$ 70. $y^5 + x^2y^3 = 1 + ye^{x^2}$
 71. $4 \cos x \sin y = 1$ 72. $1 + x = \sin(xy^2)$
 73. $e^{x/y} = x - y$ 74. $\tan(x - y) = \frac{y}{1 + x^2}$
 75. $e^y \cos x = 1 + \sin(xy)$ 76. $\sin x + \cos y = \sin x \cos y$

77–80 Use implicit differentiation to find an equation of the tangent line to the curve at the given point.

77. $x^2 + xy + y^2 = 3$, $(1, 1)$ (ellipse)
 78. $x^2 + 2xy - y^2 + x = 2$, $(1, 2)$ (hyperbola)
 79. $x^2 + y^2 = (2x^2 + 2y^2 - x)^2$, $(0, \frac{1}{2})$ (cardioid)



80. $x^{2/3} + y^{2/3} = 4$, $(-3\sqrt{3}, 1)$ (astroid)



81. Infectious disease outbreak size In Example 13 we used the equation

$$\rho e^{-qA} = 1 - A$$

to determine the rate of increase of the outbreak size A with respect to the transmissibility q . Use this same equation to find the rate of change of A with respect to ρ , the fraction of the population that is initially susceptible to infection.

82. The logistic difference equation with migration is of the form

$$N_{t+1} = N_t + N_t(1 - N_t) + m$$

where N_t is the population at time t and m is the migration rate. Suppose that as $t \rightarrow \infty$ the population size approaches a limiting value N .

- (a) What equation does N satisfy?
 (b) Use implicit differentiation to find the rate of change of N with respect to m .
 (c) Find an explicit expression for N as a function of m , differentiate it, and compare with your answer to part (b).
 83. If V is the volume of a cube with edge length x and the cube expands as time passes, find dV/dt in terms of dx/dt .
 84. (a) If A is the area of a circle with radius r and the circle expands as time passes, find dA/dt in terms of dr/dt .
 (b) Suppose oil spills from a ruptured tanker and spreads in a circular pattern. If the radius of the oil spill increases at a constant rate of 1 m/s, how fast is the area of the spill increasing when the radius is 30 m?

85. Each side of a square is increasing at a rate of 6 cm/s. At what rate is the area of the square increasing when the area of the square is 16 cm²?

86. The length of a rectangle is increasing at a rate of 8 cm/s and its width is increasing at a rate of 3 cm/s. When the length is 20 cm and the width is 10 cm, how fast is the area of the rectangle increasing?

87. Boyle's Law states that when a sample of gas is compressed at a constant temperature, the pressure P and volume V satisfy the equation $PV = C$, where C is a constant. Suppose that at a certain instant the volume is 600 cm³, the pressure is 150 kPa, and the pressure is increasing at a rate of 20 kPa/min. At what rate is the volume decreasing at this instant?

88. When air expands adiabatically (without gaining or losing heat), its pressure P and volume V are related by the equation $PV^{1.4} = C$, where C is a constant. Suppose that at a certain instant the volume is 400 cm³ and the pressure is 80 kPa and is decreasing at a rate of 10 kPa/min. At what rate is the volume increasing at this instant?

89. Bone mass In Example 1.1.6 we found an expression for the mass m of a human femur of length L in terms of the outer radius r , the inner radius r_{in} , and their ratio $k = r_{in}/r$. More generally, if the bone density is ρ , measured in g/cm³, then bone mass is given by the equation

$$m = \pi r^2 L [\rho - (\rho - 1)k^2]$$

It may happen that both ρ and k change with age, t .

- (a) If ρ changes during aging, find an expression for the rate of change of m with respect to t .
 (b) If k changes during aging, find an expression for the rate of change of m with respect to t .

90. The von Bertalanffy growth function

$$L(t) = L_{\infty} - (L_{\infty} - L_0)e^{-kt}$$

where k is a positive constant, models the length L of a fish as a function of t , the age of the fish. Here L_0 is the length at birth and L_{∞} is the final length. Suppose that the mass of a fish of length L is given by $M = aL^3$, where a is a positive constant. Calculate the rate of change of mass with respect to age.

- 91. Habitat fragmentation and species conservation** The size of a class-structured population is modeled in Section 8.8. In certain situations the long-term growth rate of a population is given by $r = \frac{1}{2}(1 + \sqrt{1 + 8s})$, where s is the annual survival probability of juveniles. Suppose this survival probability is related to habitat area a by a function $s(a)$. Determine an expression for the rate of change of growth rate with respect to a change in habitat area.

- 92. Blood flow** In Example 3.3.9 we discussed Poiseuille's law of laminar flow:

$$v = \frac{P}{4\eta l}(R^2 - r^2)$$

where v is the blood velocity at a distance r from the center of a blood vessel (a vein or artery) in the shape of a tube with radius R and length l , P is the pressure difference between the ends of the tube, and η is the viscosity of the blood. In very cold weather, blood vessels in the hands and feet contract. Suppose that a blood vessel with $l = 1$ cm, $P = 1500$ dynes/cm², $\eta = 0.027$, and $R = 0.01$ cm contracts so that $R'(t) = -0.0005$ cm/min at a particular moment. Calculate dv/dt , the rate of change of the blood flow, at the center of the blood vessel at that time.

- 93. Angiotensin-converting enzyme (ACE) inhibitors** are a type of blood pressure medication that reduces blood pressure by dilating blood vessels. Suppose that the radius R of a blood vessel is related to the dosage x of the medication by the function $R(x)$. One version of Poiseuille's Law gives the relationship between the blood pressure P and the radius as $P = 4\eta l v / R^2$, where v is the blood velocity at the center of the vessel, η is the viscosity of the blood, and l is the length of the blood vessel. Determine the rate of change of blood pressure with respect to dosage.

- 94. Brain size in fish** Brain weight B as a function of body weight W in fish has been modeled by the power function $B = 0.007W^{2/3}$, where B and W are measured in grams. A model for body weight as a function of body length L (measured in centimeters) is $W = 0.12L^{2.53}$. If, over 10 million years, the average length of a certain species of fish evolved from 15 cm to 20 cm at a constant rate, how fast was this species' brain weight increasing when the average length was 18 cm?

- 95.** Use the Chain Rule to show that if θ is measured in degrees, then

$$\frac{d}{d\theta}(\sin \theta) = \frac{\pi}{180} \cos \theta$$

(This gives one reason for the convention that radian measure is always used when dealing with trigonometric functions in calculus: The differentiation formulas would not be as simple if we used degree measure.)

- 96.** If $y = f(u)$ and $u = g(x)$, where f and g are twice differentiable functions, show that

$$\frac{d^2y}{dx^2} = \frac{d^2y}{du^2} \left(\frac{du}{dx} \right)^2 + \frac{dy}{du} \frac{d^2u}{dx^2}$$

3.6 Exponential Growth and Decay

In many natural phenomena, quantities grow or decay at a rate proportional to their size. For instance, if $y = f(t)$ is the number of individuals in a population of animals or bacteria at time t , then it seems reasonable to expect that the rate of growth $f'(t)$ is proportional to the population $f(t)$; that is, $f'(t) = kf(t)$ for some constant k . Why is this reasonable? Suppose we have a population (of bacteria, for instance) with size $P = 1000$ and at a certain time it is growing at a rate of $P' = 300$ bacteria per hour. Now let's take another 1000 bacteria of the same type and put them with the first population. Each half of the new population was growing at a rate of 300 bacteria per hour. We would expect the total population of 2000 to increase at a rate of 600 bacteria per hour initially (provided there's enough room and nutrition). So if we double the size, we double the growth rate. In general, it seems reasonable that the growth rate should be proportional to the size.

Indeed, under ideal conditions (unlimited environment, adequate nutrition, immunity to disease) the mathematical model given by the equation $f'(t) = kf(t)$ predicts what

actually happens fairly accurately. Another example occurs in nuclear physics where the mass of a radioactive substance decays at a rate proportional to the mass. In chemistry, the rate of a unimolecular first-order reaction is proportional to the concentration of the substance.

In general, if $y(t)$ is the value of a quantity y at time t and if the rate of change of y with respect to t is proportional to its size $y(t)$ at any time, then

(1)

$$\frac{dy}{dt} = ky$$

where k is a constant. Equation 1 is sometimes called the **law of natural growth** (if $k > 0$) or the **law of natural decay** (if $k < 0$). It is called a **differential equation** because it involves an unknown function y and its derivative dy/dt .

It's not hard to think of a solution of Equation 1. This equation asks us to find a function whose derivative is a constant multiple of itself. We have met such functions in this chapter. Any exponential function of the form $y(t) = Ce^{kt}$, where C is a constant, satisfies

$$y'(t) = C(ke^{kt}) = k(Ce^{kt}) = ky(t)$$

We will see in Section 7.4 that *any* function that satisfies $dy/dt = ky$ must be of the form $y = Ce^{kt}$. To see the significance of the constant C , we observe that

$$y(0) = Ce^{k \cdot 0} = C$$

Therefore C is the initial value of the function.

(2) Theorem The only solutions of the differential equation $dy/dt = ky$ are the exponential functions

$$y(t) = y(0)e^{kt}$$

■ Population Growth

What is the significance of the proportionality constant k ? In the context of population growth, where $P(t)$ is the size of a population at time t , we can write

$$(3) \quad \frac{dP}{dt} = kP \quad \text{or} \quad \frac{1}{P} \frac{dP}{dt} = k$$

The quantity

$$\frac{1}{P} \frac{dP}{dt}$$

is the growth rate divided by the population size; it is called the **relative growth rate** or **per capita growth rate**. According to (3), instead of saying “the growth rate is proportional to population size” we could say “the relative growth rate is constant.” Then (2) says that a population with constant relative growth rate must grow exponentially. Notice that the relative growth rate k appears as the coefficient of t in the exponential function Ce^{kt} . For instance, if

$$\frac{dP}{dt} = 0.02P$$

and t is measured in years, then the relative growth rate is $k = 0.02$ and the population grows at a relative rate of 2% per year. If the population at time 0 is P_0 , then the expression for the population is

$$P(t) = P_0 e^{0.02t}$$

EXAMPLE 1 | Use the fact that the world population was 2560 million in 1950 and 3040 million in 1960 to model the population of the world in the second half of the 20th century. (Assume that the growth rate is proportional to the population size.) What is the relative growth rate? Use the model to estimate the population in 1993. At what rate was the population increasing in 1993? According to the model, what will the population be in the year 2020?

SOLUTION We measure the time t in years and let $t = 0$ in the year 1950. We measure the population $P(t)$ in millions of people. Then $P(0) = 2560$ and $P(10) = 3040$. Since we are assuming that $dP/dt = kP$, Theorem 2 gives

$$P(t) = P(0)e^{kt} = 2560e^{kt}$$

$$P(10) = 2560e^{10k} = 3040$$

$$k = \frac{1}{10} \ln \frac{3040}{2560} \approx 0.017185$$

The relative growth rate is about 1.7% per year and the model is

$$P(t) = 2560e^{0.017185t}$$

We estimate that the world population in 1993 was

$$P(43) = 2560e^{0.017185(43)} \approx 5360 \text{ million}$$

At that time the rate of increase was

$$P'(43) = kP(43) \approx 92 \text{ million people per year}$$

The model predicts that the population in 2020 will be

$$P(70) = 2560e^{0.017185(70)} \approx 8524 \text{ million}$$

The graph in Figure 1 shows that the model is fairly accurate to the end of the 20th century (the dots represent the actual population), so the estimate for 1993 is quite reliable. But the prediction for 2020 is riskier.

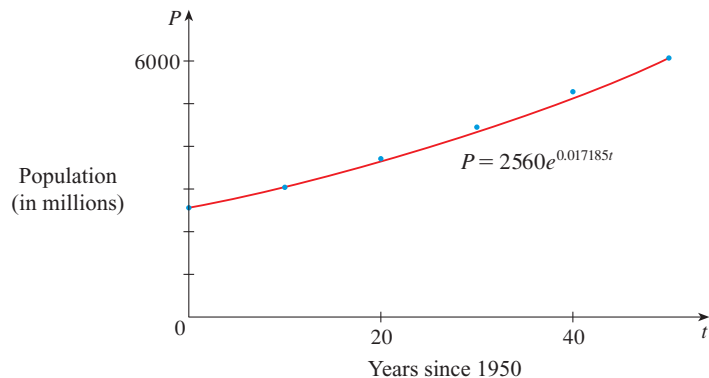


FIGURE 1

A model for world population growth in the second half of the 20th century

Radioactive Decay

Radioactive substances decay by spontaneously emitting radiation. If $m(t)$ is the mass remaining from an initial mass m_0 of the substance after time t , then the relative decay rate

$$-\frac{1}{m} \frac{dm}{dt}$$

has been found experimentally to be constant. (Since dm/dt is negative, the relative decay rate is positive.) It follows that

$$\frac{dm}{dt} = km$$

where k is a negative constant. In other words, radioactive substances decay at a rate proportional to the remaining mass. This means that we can use (2) to show that the mass decays exponentially:

$$m(t) = m_0 e^{kt}$$

Physicists express the rate of decay in terms of **half-life**, the time required for half of any given quantity to decay.

EXAMPLE 2 | The half-life of radium-226 is 1590 years.

- A sample of radium-226 has a mass of 100 mg. Find a formula for the mass of the sample that remains after t years.
- Find the mass after 1000 years correct to the nearest milligram.
- When will the mass be reduced to 30 mg?

SOLUTION

- Let $m(t)$ be the mass of radium-226 (in milligrams) that remains after t years. Then $dm/dt = km$ and $m(0) = 100$, so (2) gives

$$m(t) = m(0)e^{kt} = 100e^{kt}$$

In order to determine the value of k , we use the fact that $m(1590) = \frac{1}{2}(100)$. Thus

$$100e^{1590k} = 50 \quad \text{so} \quad e^{1590k} = \frac{1}{2}$$

and
$$1590k = \ln \frac{1}{2} = -\ln 2$$

$$k = -\frac{\ln 2}{1590}$$

Therefore
$$m(t) = 100e^{-(\ln 2)t/1590}$$

We could use the fact that $e^{\ln 2} = 2$ to write the expression for $m(t)$ in the alternative form

$$m(t) = 100 \times 2^{-t/1590}$$

- The mass after 1000 years is

$$m(1000) = 100e^{-(\ln 2)1000/1590} \approx 65 \text{ mg}$$

- We want to find the value of t such that $m(t) = 30$, that is,

$$100e^{-(\ln 2)t/1590} = 30 \quad \text{or} \quad e^{-(\ln 2)t/1590} = 0.3$$

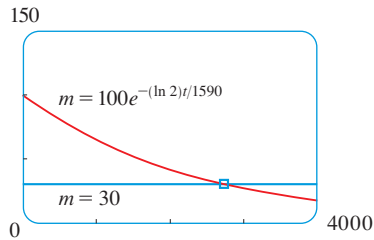


FIGURE 2

We solve this equation for t by taking the natural logarithm of both sides:

$$-\frac{\ln 2}{1590}t = \ln 0.3$$

Thus

$$t = -1590 \frac{\ln 0.3}{\ln 2} \approx 2762 \text{ years}$$

As a check on our work in Example 2, we use a calculator to draw the graph of $m(t)$ in Figure 2 together with the horizontal line $m = 30$. These curves intersect when $t \approx 2800$, and this agrees with the answer to part (c).

■ Newton's Law of Cooling

Newton's Law of Cooling states that the rate of cooling of an object is proportional to the temperature difference between the object and its surroundings, provided that this difference is not too large. (This law also applies to warming.) If we let $T(t)$ be the temperature of the object at time t and T_s be the temperature of the surroundings, then we can formulate Newton's Law of Cooling as a differential equation:

$$\frac{dT}{dt} = k(T - T_s)$$

where k is a constant. This equation is not quite the same as Equation 1, so we make the change of variable $y(t) = T(t) - T_s$. Because T_s is constant, we have $y'(t) = T'(t)$ and so the equation becomes

$$\frac{dy}{dt} = ky$$

We can then use (2) to find an expression for y , from which we can find T .

EXAMPLE 3 | A bottle of soda pop at room temperature (72°F) is placed in a refrigerator where the temperature is 44°F . After half an hour the soda pop has cooled to 61°F .

- (a) What is the temperature of the soda pop after another half hour?
- (b) How long does it take for the soda pop to cool to 50°F ?

SOLUTION

(a) Let $T(t)$ be the temperature of the soda after t minutes. The surrounding temperature is $T_s = 44^\circ\text{F}$, so Newton's Law of Cooling states that

$$\frac{dT}{dt} = k(T - 44)$$

If we let $y = T - 44$, then $y(0) = T(0) - 44 = 72 - 44 = 28$, so y satisfies

$$\frac{dy}{dt} = ky \quad y(0) = 28$$

and by (2) we have

$$y(t) = y(0)e^{kt} = 28e^{kt}$$

We are given that $T(30) = 61$, so $y(30) = 61 - 44 = 17$ and

$$28e^{30k} = 17 \quad e^{30k} = \frac{17}{28}$$

Taking logarithms, we have

$$k = \frac{\ln\left(\frac{17}{28}\right)}{30} \approx -0.01663$$

Thus

$$y(t) = 28e^{-0.01663t}$$

$$T(t) = 44 + 28e^{-0.01663t}$$

$$T(60) = 44 + 28e^{-0.01663(60)} \approx 54.3$$

So after another half hour the pop has cooled to about 54°F.

(b) We have $T(t) = 50$ when

$$44 + 28e^{-0.01663t} = 50$$

$$e^{-0.01663t} = \frac{6}{28}$$

$$t = \frac{\ln\left(\frac{6}{28}\right)}{-0.01663} \approx 92.6$$

The pop cools to 50°F after about 1 hour 33 minutes.

Notice that in Example 3, we have

$$\lim_{t \rightarrow \infty} T(t) = \lim_{t \rightarrow \infty} (44 + 28e^{-0.01663t}) = 44 + 28 \cdot 0 = 44$$

which is to be expected. The graph of the temperature function is shown in Figure 3.

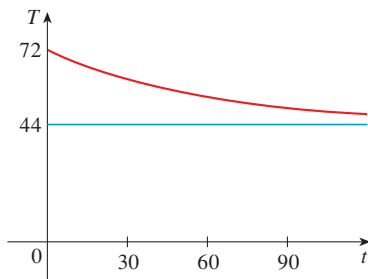


FIGURE 3

EXERCISES 3.6

1. Protozoan population A population of protozoa develops with a constant relative growth rate of 0.7944 per member per day. On day zero the population consists of two members. Find the population size after six days.

2. *E. coli* population A common inhabitant of human intestines is the bacterium *Escherichia coli*. A cell of this bacterium in a nutrient-broth medium divides into two cells every 20 minutes. The initial population of a culture is 60 cells.

- Find the relative growth rate.
- Find an expression for the number of cells after t hours.
- Find the number of cells after 8 hours.
- Find the rate of growth after 8 hours.
- When will the population reach 20,000 cells?

3. Bacteria population A bacteria culture initially contains 100 cells and grows at a rate proportional to its size. After an hour the population has increased to 420.

- Find an expression for the number of bacteria after t hours.
- Find the number of bacteria after 3 hours.
- Find the rate of growth after 3 hours.
- When will the population reach 10,000?

4. Bacteria population A bacteria culture grows with constant relative growth rate. The bacteria count was 400 after 2 hours and 25,600 after 6 hours.

- What is the relative growth rate? Express your answer as a percentage.
- What was the initial size of the culture?
- Find an expression for the number of bacteria after t hours.
- Find the number of cells after 4.5 hours.
- Find the rate of growth after 4.5 hours.
- When will the population reach 50,000?

5. World population The table gives estimates of the world population, in millions, from 1750 to 2000.

Year	Population	Year	Population
1750	790	1900	1650
1800	980	1950	2560
1850	1260	2000	6080

- Use the exponential model and the population figures for 1750 and 1800 to predict the world population in 1900 and 1950. Compare with the actual figures.

- (b) Use the exponential model and the population figures for 1850 and 1900 to predict the world population in 1950. Compare with the actual population.
- (c) Use the exponential model and the population figures for 1900 and 1950 to predict the world population in 2000. Compare with the actual population and try to explain the discrepancy.

6. Indonesian population The table gives the population of Indonesia, in millions, for the second half of the 20th century.

Year	Population
1950	83
1960	100
1970	122
1980	150
1990	182
2000	214

- (a) Assuming the population grows at a rate proportional to its size, use the census figures for 1950 and 1960 to predict the population in 1980. Compare with the actual figure.
- (b) Use the census figures for 1960 and 1980 to predict the population in 2000. Compare with the actual population.
- (c) Use the census figures for 1980 and 2000 to predict the population in 2010 and compare with the actual population of 243 million.
- (d) Use the model in part (c) to predict the population in 2020. Do you think the prediction will be too high or too low? Why?
- 7.** The half-life of cesium-137 is 30 years. Suppose we have a 100-mg sample.
- Find the mass that remains after t years.
 - How much of the sample remains after 100 years?
 - What is the rate of decay after 100 years?
 - After how long will only 1 mg remain?
- 8.** Strontium-90 has a half-life of 28 days.
- A sample has a mass of 50 mg initially. Find a formula for the mass remaining after t days.
 - Find the mass remaining after 40 days.
 - What is the rate of decay after 40 days?
 - How long does it take the sample to decay to a mass of 2 mg?
 - Sketch the graph of the mass function.
- 9.** A sample of tritium-3 decayed to 94.5% of its original amount after a year.
- What is the half-life of tritium-3?
 - How long would it take the sample to decay to 20% of its original amount?

10. Radiometric dating Scientists can determine the age of ancient objects by the method of *radiometric dating*. The bombardment of the upper atmosphere by cosmic rays converts nitrogen to a radioactive isotope of carbon, ^{14}C , with a half-life of about 5730 years. Vegetation absorbs carbon dioxide through the atmosphere and animal life assimilates ^{14}C through food chains. When a plant or animal dies, it stops replacing its carbon and the amount of ^{14}C begins to decrease through radioactive decay. Therefore the level of radioactivity must also decay exponentially.

A discovery revealed a parchment fragment that had about 74% as much ^{14}C radioactivity as does plant material on the earth today. Estimate the age of the parchment.

11. Dating dinosaurs Dinosaur fossils are too old to be reliably dated using carbon-14, which has a half-life of about 5730 years. (See Exercise 10.) Suppose we had a 68-million-year-old dinosaur fossil. What fraction of the living dinosaur's ^{14}C would be remaining today? Suppose the minimum detectable amount is 0.1%. What is the maximum age of a fossil that we could date using ^{14}C ?



- 12. Dating dinosaurs with potassium** Dinosaur fossils are often dated by using an element other than carbon, such as potassium-40, that has a longer half-life (in this case, approximately 1.25 billion years). Suppose the minimum detectable amount is 0.1% and a dinosaur is dated with ^{40}K to be 68 million years old. Is such a dating possible? In other words, what is the maximum age of a fossil that we could date using ^{40}K ?
- 13.** A roast turkey is taken from an oven when its temperature has reached 185°F and is placed on a table in a room where the temperature is 75°F .
- If the temperature of the turkey is 150°F after half an hour, what is the temperature after 45 minutes?
 - When will the turkey have cooled to 100°F ?
- 14.** In a murder investigation, the temperature of the corpse was 32.5°C at 1:30 PM and 30.3°C an hour later. Normal body temperature is 37.0°C and the temperature of the surroundings was 20.0°C . When did the murder take place?

15. When a cold drink is taken from a refrigerator, its temperature is 5°C . After 25 minutes in a 20°C room its temperature has increased to 10°C .
 - (a) What is the temperature of the drink after 50 minutes?
 - (b) When will its temperature be 15°C ?
16. A freshly brewed cup of coffee has temperature 95°C in a 20°C room. When its temperature is 70°C , it is cooling at a rate of 1°C per minute. When does this occur?
17. The rate of change of atmospheric pressure P with respect to altitude h is proportional to P , provided that the temperature is constant. At 15°C the pressure is 101.3 kPa at sea level and 87.14 kPa at $h = 1000$ m.
 - (a) What is the pressure at an altitude of 3000 m?
 - (b) What is the pressure at the top of Mount McKinley, at an altitude of 6187 m?

PROJECT Controlling Red Blood Cell Loss During Surgery



A typical volume of blood in the human body is about 5 L. A certain percentage of that volume (called the *hematocrit*) consists of red blood cells (RBCs); typically the hematocrit is about 45% in males. Suppose that a surgery takes four hours and a male patient bleeds 2 L of blood. During surgery the patient's blood volume is maintained at 5 L by injection of saline solution, which mixes quickly with the blood but dilutes it so that the hematocrit decreases as time passes.

1. Assuming that the rate of RBC loss is proportional to the concentration of RBCs, determine the patient's concentration of RBCs by the end of the operation.
2. A procedure called *acute normovolemic hemodilution* (ANH) has been developed to minimize RBC loss during surgery. In this procedure blood is extracted from the patient before the operation and replaced with saline solution. This dilutes the patient's blood, resulting in fewer RBCs being lost during the bleeding. The extracted blood is then returned to the patient after surgery. Only a certain amount of blood can be extracted, however, because the RBC concentration can never be allowed to drop below 25% during surgery. What is the maximum amount of blood that can be extracted in the ANH procedure for the surgery described in this project?
3. What is the RBC loss without the ANH procedure? What is the loss if the procedure is carried out with the volume calculated in Problem 2?

3.7 Derivatives of the Logarithmic and Inverse Tangent Functions

In this section we use implicit differentiation to find the derivatives of the logarithmic functions $y = \log_b x$ and, in particular, the natural logarithmic function $y = \ln x$, as well as the inverse tangent function $y = \tan^{-1}x$.

Differentiating Logarithmic Functions

It can be proved that logarithmic functions are differentiable. This is certainly plausible from their graphs (see Figure 1.5.12). We now differentiate the general logarithmic function $y = \log_b x$ by using the fact that it is the inverse function of the exponential function with base b .

(1)

$$\frac{d}{dx}(\log_b x) = \frac{1}{x \ln b}$$

PROOF Let $y = \log_b x$. Then

$$b^y = x$$

Formula 3.5.5 says that

$$\frac{d}{dx}(b^x) = b^x \ln b$$

Differentiating this equation implicitly with respect to x , using Formula 3.5.5, we get

$$b^y(\ln b) \frac{dy}{dx} = 1$$

and so

$$\frac{dy}{dx} = \frac{1}{b^y \ln b} = \frac{1}{x \ln b}$$

If we put $b = e$ in Formula 1, then the factor $\ln b$ on the right side becomes $\ln e = 1$ and we get the formula for the derivative of the natural logarithmic function $\log_e x = \ln x$:

(2)

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

By comparing Formulas 1 and 2, we see one of the main reasons that natural logarithms (logarithms with base e) are used in calculus: The differentiation formula is simplest when $b = e$ because $\ln e = 1$.

EXAMPLE 1 | Differentiate $y = \ln(x^3 + 1)$.**SOLUTION** To use the Chain Rule, we let $u = x^3 + 1$. Then $y = \ln u$, so

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} = \frac{1}{u} \frac{du}{dx} \\ &= \frac{1}{x^3 + 1} (3x^2) = \frac{3x^2}{x^3 + 1} \end{aligned}$$

In general, if we combine Formula 2 with the Chain Rule as in Example 1, we get

(3)

$$\frac{d}{dx}(\ln u) = \frac{1}{u} \frac{du}{dx}$$

or

$$\frac{d}{dx}[\ln g(x)] = \frac{g'(x)}{g(x)}$$

EXAMPLE 2 | Find $\frac{d}{dx} \ln(\sin x)$.**SOLUTION** Using (3), we have

$$\frac{d}{dx} \ln(\sin x) = \frac{1}{\sin x} \frac{d}{dx}(\sin x) = \frac{1}{\sin x} \cos x = \cot x$$

EXAMPLE 3 | Differentiate $f(x) = \sqrt{\ln x}$.**SOLUTION** This time the logarithm is the inner function, so the Chain Rule gives

$$f'(x) = \frac{1}{2}(\ln x)^{-1/2} \frac{d}{dx}(\ln x) = \frac{1}{2\sqrt{\ln x}} \cdot \frac{1}{x} = \frac{1}{2x\sqrt{\ln x}}$$

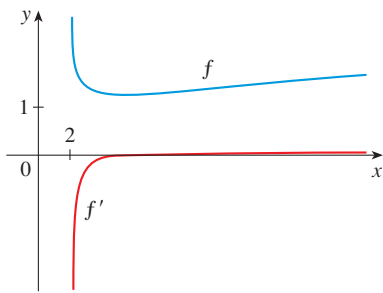
EXAMPLE 4 | Differentiate $f(x) = \log_{10}(2 + \sin x)$.**SOLUTION** Using Formula 1 with $b = 10$, we have

$$\begin{aligned} f'(x) &= \frac{d}{dx} \log_{10}(2 + \sin x) \\ &= \frac{1}{(2 + \sin x) \ln 10} \frac{d}{dx}(2 + \sin x) \\ &= \frac{\cos x}{(2 + \sin x) \ln 10} \end{aligned}$$

EXAMPLE 5 | Find $\frac{d}{dx} \ln\left(\frac{x+1}{\sqrt{x-2}}\right)$.**SOLUTION 1**

$$\begin{aligned} \frac{d}{dx} \ln\left(\frac{x+1}{\sqrt{x-2}}\right) &= \frac{1}{\frac{x+1}{\sqrt{x-2}}} \frac{d}{dx}\left(\frac{x+1}{\sqrt{x-2}}\right) \\ &= \frac{\sqrt{x-2}}{x+1} \frac{\sqrt{x-2} \cdot 1 - (x+1)\left(\frac{1}{2}\right)(x-2)^{-1/2}}{x-2} \\ &= \frac{x-2 - \frac{1}{2}(x+1)}{(x+1)(x-2)} \\ &= \frac{x-5}{2(x+1)(x-2)} \end{aligned}$$

Figure 1 shows the graph of the function f of Example 5 together with the graph of its derivative. It gives a visual check on our calculation. Notice that $f'(x)$ is large negative when f is rapidly decreasing.

**FIGURE 1****SOLUTION 2** If we first simplify the given function using the laws of logarithms, then the differentiation becomes easier:

$$\begin{aligned} \frac{d}{dx} \ln\left(\frac{x+1}{\sqrt{x-2}}\right) &= \frac{d}{dx} \left[\ln(x+1) - \frac{1}{2} \ln(x-2) \right] \\ &= \frac{1}{x+1} - \frac{1}{2} \left(\frac{1}{x-2} \right) \end{aligned}$$

(This answer can be left as written, but if we used a common denominator we would see that it gives the same answer as in Solution 1.)

Figure 2 shows the graph of the function $f(x) = \ln|x|$ in Example 6 and its derivative $f'(x) = 1/x$. Notice that when x is small, the graph of $y = \ln|x|$ is steep and so $f'(x)$ is large (positive or negative).

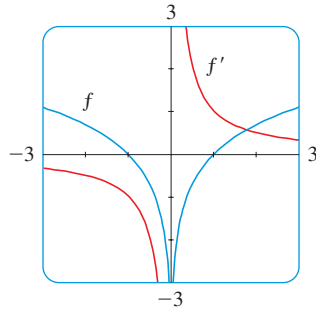


FIGURE 2

EXAMPLE 6 | Find $f'(x)$ if $f(x) = \ln|x|$.

SOLUTION Since

$$f(x) = \begin{cases} \ln x & \text{if } x > 0 \\ \ln(-x) & \text{if } x < 0 \end{cases}$$

it follows that

$$f'(x) = \begin{cases} \frac{1}{x} & \text{if } x > 0 \\ \frac{1}{-x}(-1) = \frac{1}{x} & \text{if } x < 0 \end{cases}$$

Thus $f'(x) = 1/x$ for all $x \neq 0$.

The result of Example 6 is worth remembering:

(4)

$$\frac{d}{dx} \ln|x| = \frac{1}{x}$$

Logarithmic Differentiation

The calculation of derivatives of complicated functions involving products, quotients, or powers can often be simplified by taking logarithms. The method used in the following example is called **logarithmic differentiation**.

EXAMPLE 7 | Differentiate $y = \frac{x^{3/4}\sqrt{x^2+1}}{(3x+2)^5}$.

SOLUTION We take logarithms of both sides of the equation and use the Laws of Logarithms to simplify:

$$\ln y = \frac{3}{4} \ln x + \frac{1}{2} \ln(x^2 + 1) - 5 \ln(3x + 2)$$

Differentiating implicitly with respect to x gives

$$\frac{1}{y} \frac{dy}{dx} = \frac{3}{4} \cdot \frac{1}{x} + \frac{1}{2} \cdot \frac{2x}{x^2 + 1} - 5 \cdot \frac{3}{3x + 2}$$

Solving for dy/dx , we get

$$\frac{dy}{dx} = y \left(\frac{3}{4x} + \frac{x}{x^2 + 1} - \frac{15}{3x + 2} \right)$$

If we hadn't used logarithmic differentiation in Example 7, we would have had to use both the Quotient Rule and the Product Rule. The resulting calculation would have been horrendous.

Because we have an explicit expression for y , we can substitute and write

$$\frac{dy}{dx} = \frac{x^{3/4}\sqrt{x^2+1}}{(3x+2)^5} \left(\frac{3}{4x} + \frac{x}{x^2+1} - \frac{15}{3x+2} \right)$$

Steps in Logarithmic Differentiation

1. Take natural logarithms of both sides of an equation $y = f(x)$ and use the Laws of Logarithms to simplify.
2. Differentiate implicitly with respect to x .
3. Solve the resulting equation for y' .

If $f(x) < 0$ for some values of x , then $\ln f(x)$ is not defined, but we can write $|y| = |f(x)|$ and use Equation 4. We illustrate this procedure by proving the general version of the Power Rule, as promised in Section 3.3.

The Power Rule If n is any real number and $f(x) = x^n$, then

$$f'(x) = nx^{n-1}$$

If $x = 0$, we can show that $f'(0) = 0$ for $n > 1$ directly from the definition of a derivative.

PROOF Let $y = x^n$ and use logarithmic differentiation:

$$\ln |y| = \ln |x|^n = n \ln |x| \quad x \neq 0$$

Therefore
$$\frac{y'}{y} = \frac{n}{x}$$

Hence
$$y' = n \frac{y}{x} = n \frac{x^n}{x} = nx^{n-1}$$
 ■

To differentiate a function of the form $y = [f(x)]^{g(x)}$, where both the base and the exponent are functions, logarithmic differentiation can be used as in the following example.

EXAMPLE 8 | Differentiate $y = x^{\sqrt{x}}$.

SOLUTION 1 Because both the base and the exponent are variable, we use logarithmic differentiation:

$$\begin{aligned} \ln y &= \ln x^{\sqrt{x}} = \sqrt{x} \ln x \\ \frac{y'}{y} &= \sqrt{x} \cdot \frac{1}{x} + (\ln x) \frac{1}{2\sqrt{x}} \\ y' &= y \left(\frac{1}{\sqrt{x}} + \frac{\ln x}{2\sqrt{x}} \right) = x^{\sqrt{x}} \left(\frac{2 + \ln x}{2\sqrt{x}} \right) \end{aligned}$$

SOLUTION 2 Another method is to write $x^{\sqrt{x}} = (e^{\ln x})^{\sqrt{x}}$:

$$\begin{aligned} \frac{d}{dx}(x^{\sqrt{x}}) &= \frac{d}{dx}(e^{\sqrt{x} \ln x}) = e^{\sqrt{x} \ln x} \frac{d}{dx}(\sqrt{x} \ln x) \\ &= x^{\sqrt{x}} \left(\frac{2 + \ln x}{2\sqrt{x}} \right) \quad (\text{as in Solution 1}) \end{aligned}$$
 ■

Figure 3 illustrates Example 8 by showing the graphs of $f(x) = x^{\sqrt{x}}$ and its derivative.

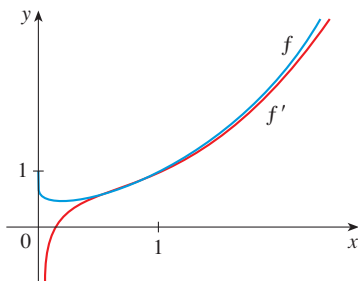


FIGURE 3

■ The Number e as a Limit

We have shown that if $f(x) = \ln x$, then $f'(x) = 1/x$. Thus $f'(1) = 1$. We now use this fact to express the number e as a limit.

From the definition of a derivative as a limit, we have

$$\begin{aligned} f'(1) &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{x \rightarrow 0} \frac{f(1+x) - f(1)}{x} \\ &= \lim_{x \rightarrow 0} \frac{\ln(1+x) - \ln 1}{x} = \lim_{x \rightarrow 0} \frac{1}{x} \ln(1+x) \\ &= \lim_{x \rightarrow 0} \ln(1+x)^{1/x} \end{aligned}$$

Because $f'(1) = 1$, we have

$$\lim_{x \rightarrow 0} \ln(1+x)^{1/x} = 1$$

Then, by Theorem 2.5.7 and the continuity of the exponential function, we have

$$e = e^1 = e^{\lim_{x \rightarrow 0} \ln(1+x)^{1/x}} = \lim_{x \rightarrow 0} e^{\ln(1+x)^{1/x}} = \lim_{x \rightarrow 0} (1+x)^{1/x}$$

(5)

$$e = \lim_{x \rightarrow 0} (1+x)^{1/x}$$

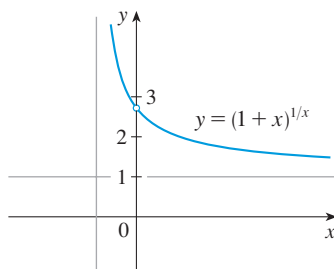


FIGURE 4

x	$(1+x)^{1/x}$
0.1	2.59374246
0.01	2.70481383
0.001	2.71692393
0.0001	2.71814593
0.00001	2.71826824
0.000001	2.71828047
0.0000001	2.71828169
0.00000001	2.71828181

Formula 5 is illustrated by the graph of the function $y = (1+x)^{1/x}$ in Figure 4 and a table of values for small values of x . This illustrates the fact that, correct to seven decimal places,

$$e \approx 2.7182818$$

If we put $n = 1/x$ in Formula 5, then $n \rightarrow \infty$ as $x \rightarrow 0^+$ and so an alternative expression for e is

(6)

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$

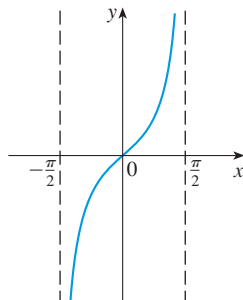


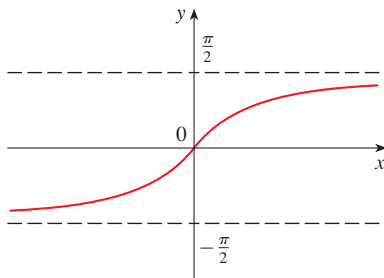
FIGURE 5

$$y = \tan x, -\frac{\pi}{2} < x < \frac{\pi}{2}$$

■ Differentiating the Inverse Tangent Function

Recall from Section 1.5 that the only functions that have inverse functions are one-to-one functions. The tangent function, however, is not one-to-one and so it doesn't have an inverse function. But we can make it one-to-one by restricting its domain to the interval $(-\pi/2, \pi/2)$. Thus the **inverse tangent function** is defined as the inverse of the function $f(x) = \tan x$, $-\pi/2 < x < \pi/2$, as shown in Figure 5. It is denoted by $\tan^{-1}$ or arctan.

$$\tan^{-1} x = y \iff \tan y = x \quad \text{and} \quad -\frac{\pi}{2} < y < \frac{\pi}{2}$$

**FIGURE 6**

$$y = \tan^{-1}x = \arctan x$$

The inverse tangent function, $\tan^{-1} = \arctan$, has domain $\mathbb{R}$ and range $(-\pi/2, \pi/2)$. Its graph is shown in Figure 6.

We know that

$$\lim_{x \rightarrow (\pi/2)^-} \tan x = \infty \quad \text{and} \quad \lim_{x \rightarrow -(\pi/2)^+} \tan x = -\infty$$

and so the lines $x = \pm\pi/2$ are vertical asymptotes of the graph of $\tan$. Since the graph of $\tan^{-1}$ is obtained by reflecting the graph of the restricted tangent function about the line $y = x$, it follows that the lines $y = \pi/2$ and $y = -\pi/2$ are horizontal asymptotes of the graph of $\tan^{-1}$. This fact is expressed by the following limits:

(7)

$$\lim_{x \rightarrow \infty} \tan^{-1}x = \frac{\pi}{2} \quad \lim_{x \rightarrow -\infty} \tan^{-1}x = -\frac{\pi}{2}$$

We can use implicit differentiation to derive the formula for the derivative of the arctangent function. If $y = \tan^{-1}x$, then $\tan y = x$. Differentiating this latter equation implicitly with respect to x , we have

$$\sec^2 y \frac{dy}{dx} = 1$$

Of all the inverse trigonometric functions, the most useful for our purposes is the inverse tangent function, as we will see in Section 5.8.

and so

$$\frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{1 + \tan^2 y} = \frac{1}{1 + x^2}$$

(8)

$$\frac{d}{dx}(\tan^{-1}x) = \frac{1}{1 + x^2}$$

EXAMPLE 9 | Differentiate:

$$(a) \ y = \frac{1}{\tan^{-1}x}$$

$$(b) \ f(x) = x \arctan \sqrt{x}$$

SOLUTION

$$\begin{aligned} (a) \quad \frac{dy}{dx} &= \frac{d}{dx}(\tan^{-1}x)^{-1} = -(\tan^{-1}x)^{-2} \frac{d}{dx}(\tan^{-1}x) \\ &= -\frac{1}{(\tan^{-1}x)^2(1 + x^2)} \end{aligned}$$

(b) Using the Product Rule and the Chain Rule, we have

$$\begin{aligned} f'(x) &= x \frac{1}{1 + (\sqrt{x})^2} \left(\frac{1}{2}x^{-1/2}\right) + \arctan \sqrt{x} \\ &= \frac{\sqrt{x}}{2(1 + x)} + \arctan \sqrt{x} \end{aligned}$$

EXERCISES 3.7

1. Explain why the natural logarithmic function $y = \ln x$ is used much more frequently in calculus than the other logarithmic functions $y = \log_b x$.

2–20 Differentiate the function.

2. $f(x) = x \ln x - x$
3. $f(x) = \sin(\ln x)$
4. $f(x) = \ln(\sin^2 x)$
5. $f(x) = \log_2(1 - 3x)$
6. $f(x) = \log_5(xe^x)$
7. $f(x) = \sqrt[5]{\ln x}$
8. $f(x) = \ln \sqrt[5]{x}$
9. $f(x) = \sin x \ln(5x)$
10. $f(t) = \frac{1 + \ln t}{1 - \ln t}$
11. $F(t) = \ln \frac{(2t + 1)^3}{(3t - 1)^4}$
12. $h(x) = \ln(x + \sqrt{x^2 - 1})$
13. $g(x) = \ln(x\sqrt{x^2 - 1})$
14. $F(y) = y \ln(1 + e^y)$
15. $y = \ln|2 - x - 5x^2|$
16. $H(z) = \ln \sqrt{\frac{a^2 - z^2}{a^2 + z^2}}$
17. $y = \ln(e^{-x} + xe^{-x})$
18. $y = [\ln(1 + e^x)]^2$
19. $y = 2x \log_{10} \sqrt{x}$
20. $y = \log_2(e^{-x} \cos \pi x)$

21–22 Find y' and y'' .

21. $y = x^2 \ln(2x)$
22. $y = \frac{\ln x}{x^2}$

23–24 Differentiate f and find the domain of f .

23. $f(x) = \frac{x}{1 - \ln(x - 1)}$
24. $f(x) = \ln \ln x$

25–26 Find an equation of the tangent line to the curve at the given point.

25. $y = \ln(x^2 - 3x + 1)$, $(3, 0)$
26. $y = x^2 \ln x$, $(1, 0)$

27. If $f(x) = \frac{\ln x}{x^2}$, find $f'(1)$.

-  **28.** Find equations of the tangent lines to the curve $y = (\ln x)/x$ at the points $(1, 0)$ and $(e, 1/e)$. Illustrate by graphing the curve and its tangent lines.

- 29. Dialysis** The project on page 458 models the removal of urea from the bloodstream via dialysis. Given that the initial urea concentration, measured in mg/mL, is c (where $c > 1$), the duration of dialysis required for certain conditions is given by the equation

$$t = \ln \left(\frac{3c + \sqrt{9c^2 - 8c}}{2} \right)$$

Calculate the derivative of t with respect to c and interpret it.

- 30. Genetic drift** A population of fruit flies contains two genetically determined kinds of individuals: white-eyed flies and red-eyed flies. Suppose that a scientist maintains the population at constant size N by randomly choosing N juvenile flies after reproduction to form the next generation. Eventually, because of the random sampling in each generation, by chance the population will contain only a single type of fly. This is called *genetic drift*. Suppose that the initial fraction of the population that are white-eyed is p_0 . An equation for the average number of generations required before all flies are white-eyed (given that this occurs instead of all flies being red-eyed) is

$$g = -2N \frac{1 - p_0}{p_0} \ln(1 - p_0)$$

Calculate the derivative of g with respect to p_0 and explain its meaning.

- 31. Carbon dating** If N is the measured amount of ^{14}C in a fossil organism and N_0 is the amount in living organisms, then the estimated age of the fossil is given by the equation

$$a = \frac{5370}{\ln 2} \ln \left(\frac{N_0}{N} \right)$$

Calculate da/dN and interpret it.

- 32.** Let $f(x) = \log_b(3x^2 - 2)$. For what value of b is $f'(1) = 3$?

33–41 Use logarithmic differentiation to find the derivative of the function.

33. $y = (2x + 1)^5(x^4 - 3)^6$
34. $y = \sqrt{x}e^{x^2}(x^2 + 1)^{10}$
35. $y = \frac{\sin^2 x \tan^4 x}{(x^2 + 1)^2}$
36. $y = \sqrt[4]{\frac{x^2 + 1}{x^2 - 1}}$
37. $y = x^x$
38. $y = x^{\cos x}$

39. $y = (\cos x)^x$

40. $y = \sqrt{x}^x$

41. $y = (\tan x)^{1/x}$

42. Predator-prey dynamics In Chapter 7 we study a model for the population sizes of a predator and its prey species. If $u(t)$ and $v(t)$ denote the prey and predator population sizes at time t , an equation relating the two is

$$ve^{-v}u^\alpha e^{-\alpha u} = c$$

where c and α are positive constants. Use logarithmic differentiation to obtain an equation relating the relative (per capita) rate of change of predator (that is, v'/v) to that of prey (that is, u'/u).

43–48 Find the derivative of the function. Simplify where possible.

43. $y = (\tan^{-1}x)^2$

44. $y = \tan^{-1}(x^2)$

45. $y = \arctan(\cos \theta)$

46. $f(x) = x \ln(\arctan x)$

47. $y = \tan^{-1}(x - \sqrt{1 + x^2})$

48. $y = \arctan \sqrt{\frac{1-x}{1+x}}$

49–50 Find the limit.

49. $\lim_{x \rightarrow \infty} \arctan(e^x)$

50. $\lim_{x \rightarrow 0^+} \tan^{-1}(\ln x)$

51. Find y' if $y = \ln(x^2 + y^2)$.

52. Find y' if $x^y = y^x$.

53. Find a formula for $f^{(n)}(x)$ if $f(x) = \ln(x - 1)$.

54. Find $\frac{d^9}{dx^9}(x^8 \ln x)$.

55. Use the definition of derivative to prove that

$$\lim_{x \rightarrow 0} \frac{\ln(1 + x)}{x} = 1$$

56. Show that $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$ for any $x > 0$.

3.8 Linear Approximations and Taylor Polynomials

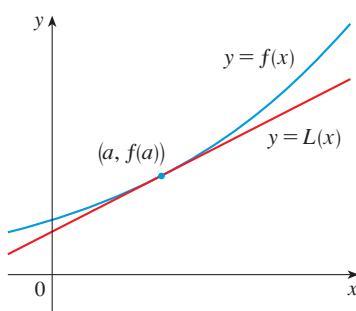


FIGURE 1

■ Tangent Line Approximations

We have seen that a curve lies very close to its tangent line near the point of tangency. In fact, by zooming in toward a point on the graph of a differentiable function, we noticed that the graph looks more and more like its tangent line. (See Figure 3.1.5.) This observation is the basis for a method of finding approximate values of functions.

The idea is that it might be easy to calculate a value $f(a)$ of a function, but difficult (or even impossible) to compute nearby values of f . So we settle for the easily computed values of the linear function L whose graph is the tangent line of f at $(a, f(a))$. (See Figure 1.)

In other words, we use the tangent line at $(a, f(a))$ as an approximation to the curve $y = f(x)$ when x is near a . An equation of this tangent line is

$$y = f(a) + f'(a)(x - a)$$

and the approximation

$$(1) \quad f(x) \approx f(a) + f'(a)(x - a)$$

is called the **linear approximation** or **tangent line approximation** of f at a . The linear function whose graph is this tangent line, that is,

$$(2) \quad L(x) = f(a) + f'(a)(x - a)$$

is called the **linearization** of f at a .

EXAMPLE 1 | Find the linearization of the function $f(x) = \sqrt{x + 3}$ at $a = 1$ and use it to approximate the numbers $\sqrt{3.98}$ and $\sqrt{4.05}$. Are these approximations overestimates or underestimates?

SOLUTION The derivative of $f(x) = (x + 3)^{1/2}$ is

$$f'(x) = \frac{1}{2}(x + 3)^{-1/2} = \frac{1}{2\sqrt{x + 3}}$$

and so we have $f(1) = 2$ and $f'(1) = \frac{1}{4}$. Putting these values into Equation 2, we see that the linearization is

$$L(x) = f(1) + f'(1)(x - 1) = 2 + \frac{1}{4}(x - 1) = \frac{7}{4} + \frac{x}{4}$$

The corresponding linear approximation (1) is

$$\sqrt{x + 3} \approx \frac{7}{4} + \frac{x}{4} \quad (\text{when } x \text{ is near } 1)$$

In particular, we have

$$\sqrt{3.98} \approx \frac{7}{4} + \frac{0.98}{4} = 1.995 \quad \text{and} \quad \sqrt{4.05} \approx \frac{7}{4} + \frac{1.05}{4} = 2.0125$$

The linear approximation is illustrated in Figure 2. We see that, indeed, the tangent line approximation is a good approximation to the given function when x is near 1. We also see that our approximations are overestimates because the tangent line lies above the curve.

Of course, a calculator could give us approximations for $\sqrt{3.98}$ and $\sqrt{4.05}$, but the linear approximation gives an approximation *over an entire interval*. ■

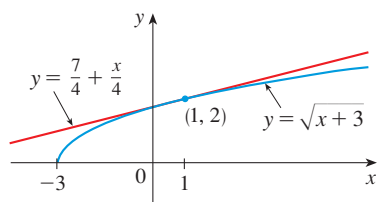


FIGURE 2

In the following table we compare the estimates from the linear approximation in Example 1 with the true values. Notice from this table, and also from Figure 2, that the tangent line approximation gives good estimates when x is close to 1 but the accuracy of the approximation deteriorates when x is farther away from 1.

	x	From $L(x)$	Actual value
$\sqrt{3.9}$	0.9	1.975	1.97484176 ...
$\sqrt{3.98}$	0.98	1.995	1.99499373 ...
$\sqrt{4}$	1	2	2.00000000 ...
$\sqrt{4.05}$	1.05	2.0125	2.01246117 ...
$\sqrt{4.1}$	1.1	2.025	2.02484567 ...
$\sqrt{5}$	2	2.25	2.23606797 ...
$\sqrt{6}$	3	2.5	2.44948974 ...

How good is the approximation that we obtained in Example 1? The next example shows that by using a graphing calculator or computer we can determine an interval throughout which a linear approximation provides a specified accuracy.

EXAMPLE 2 | For what values of x is the linear approximation

$$\sqrt{x+3} \approx \frac{7}{4} + \frac{x}{4}$$

accurate to within 0.5? What about accuracy to within 0.1?

SOLUTION Accuracy to within 0.5 means that the functions should differ by less than 0.5:

$$\left| \sqrt{x+3} - \left(\frac{7}{4} + \frac{x}{4} \right) \right| < 0.5$$

Equivalently, we could write

$$\sqrt{x+3} - 0.5 < \frac{7}{4} + \frac{x}{4} < \sqrt{x+3} + 0.5$$

This says that the linear approximation should lie between the curves obtained by shifting the curve $y = \sqrt{x+3}$ upward and downward by an amount 0.5. Figure 3 shows the tangent line $y = (7+x)/4$ intersecting the upper curve $y = \sqrt{x+3} + 0.5$ at P and Q . Zooming in and using the cursor, we estimate that the x -coordinate of P is about -2.66 and the x -coordinate of Q is about 8.66 . Thus we see from the graph that the approximation

$$\sqrt{x+3} \approx \frac{7}{4} + \frac{x}{4}$$

is accurate to within 0.5 when $-2.6 < x < 8.6$. (We have rounded to a narrower interval to be safe.)

Similarly, from Figure 4 we see that the approximation is accurate to within 0.1 when $-1.1 < x < 3.9$. ■

EXAMPLE 3 | **Population growth** If $N(t)$ represents a population size at time t and the rate of growth as a function of N is $f(N)$, what is the linear approximation of the growth rate at $N = 0$?**SOLUTION** The growth rate is

$$\frac{dN}{dt} = f(N)$$

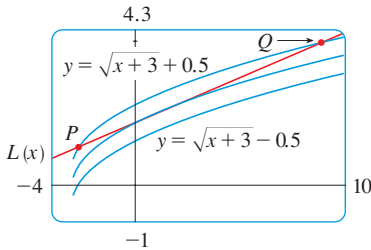
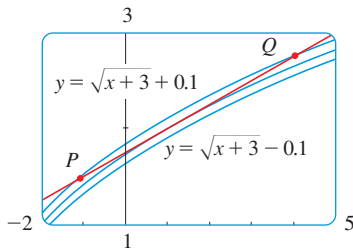
The linearization of $f(N)$ at $N = 0$ is

$$L(N) = f(0) + f'(0)N$$

We can assume that $f(0) = 0$ because when the population has size $N = 0$, its growth rate will be zero. So $L(N) = f'(0)N$. If we let the initial growth rate be $f'(0) = r$, then the linear approximation is

$$\frac{dN}{dt} \approx rN$$

This means that for small population sizes, the population grows approximately exponentially. (Recall from Theorem 3.6.2 that the only solutions of the equation $dN/dt = rN$ are exponential functions.) ■

EXAMPLE 4 | Find the linear approximation of the sine function at 0.**FIGURE 3****FIGURE 4**

SOLUTION If we let $f(x) = \sin x$, then $f'(x) = \cos x$ and so the linearization at 0 is

$$L(x) = f(0) + f'(0)(x - 0) = \sin 0 + (\cos 0)(x) = x$$

So the linear approximation at 0 is

$$\sin x \approx x$$

This approximation is used in optics when x is small: the results of calculations made with this linear approximation became the basic theoretical tool used to design lenses.

■ Newton's Method

How would you solve an equation like $\cos x = x$? Aside from linear and quadratic equations, most equations don't have simple formulas for their roots. Many calculators have numerical rootfinders that enable us to find approximate roots of equations, though they need to be used with care.

How do those numerical rootfinders work? They use a variety of methods, but most of them make some use of **Newton's method**, also called the **Newton-Raphson method**. We will explain how this method works, partly to show what happens inside a calculator or computer, and partly as an application of the idea of linear approximation.

The geometry behind Newton's method is shown in Figure 5, where the root that we are trying to find is labeled r . We start with a first approximation x_1 , which is obtained by guessing, or from a rough sketch of the graph of f , or from a computer-generated graph of f . Consider the tangent line L to the curve $y = f(x)$ at the point $(x_1, f(x_1))$ and look at the x -intercept of L , labeled x_2 . The idea behind Newton's method is that the tangent line is close to the curve and so its x -intercept, x_2 , is close to the x -intercept of the curve (namely, the root r that we are seeking). Because the tangent is a line, we can easily find its x -intercept.

To find a formula for x_2 in terms of x_1 we use the fact that the slope of L is $f'(x_1)$, so its equation is

$$y - f(x_1) = f'(x_1)(x - x_1)$$

Since the x -intercept of L is x_2 , we set $y = 0$ and obtain

$$0 - f(x_1) = f'(x_1)(x_2 - x_1)$$

If $f'(x_1) \neq 0$, we can solve this equation for x_2 :

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

We use x_2 as a second approximation to r .

Next we repeat this procedure with x_1 replaced by the second approximation x_2 , using the tangent line at $(x_2, f(x_2))$. This gives a third approximation:

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

If we keep repeating this process, we obtain a sequence of approximations $x_1, x_2, x_3, x_4, \dots$ as shown in Figure 6. In general, if the n th approximation is x_n and $f'(x_n) \neq 0$, then the next approximation is given by

(3)

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

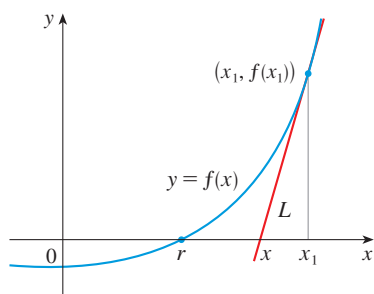


FIGURE 5

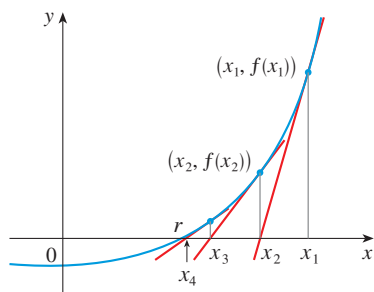


FIGURE 6

Limits of sequences were defined in Section 2.1.

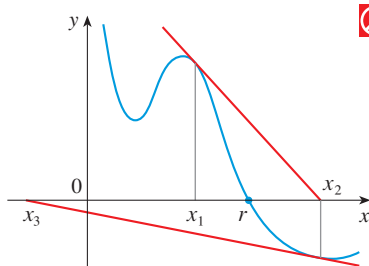


FIGURE 7

TEC In Module 3.8 you can investigate how Newton's Method works for several functions and what happens when you change x_1 .

Figure 8 shows the geometry behind the first step in Newton's method in Example 5. Since $f'(2) = 10$, the tangent line to $y = x^3 - 2x - 5$ at $(2, -1)$ has equation $y = 10x - 21$ and so its x -intercept is $x_2 = 2.1$.

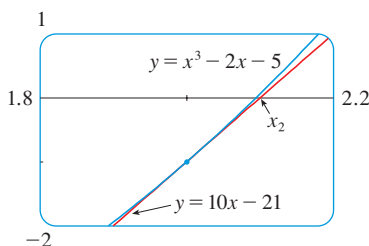


FIGURE 8

If the numbers x_n approach the desired root r as n becomes large, that is,

$$\lim_{n \rightarrow \infty} x_n = r$$

then we use them as approximations to r . Although the sequence of successive approximations converges to the desired root for functions of the type illustrated in Figure 6, in certain circumstances the sequence may not converge. For example, consider the situation shown in Figure 7. You can see that x_2 is a worse approximation than x_1 . This is likely to be the case when $f'(x_1)$ is close to 0. It might even happen that an approximation (such as x_3 in Figure 7) falls outside the domain of f . **Then Newton's method fails and a better initial approximation x_1 should be chosen.** See Exercises 33–34 for specific examples in which Newton's method works very slowly or does not work at all.

EXAMPLE 5 | Starting with $x_1 = 2$, find the third approximation x_3 to the root of the equation $x^3 - 2x - 5 = 0$.

SOLUTION We apply Newton's method with

$$f(x) = x^3 - 2x - 5 \quad \text{and} \quad f'(x) = 3x^2 - 2$$

Newton himself used this equation to illustrate his method and he chose $x_1 = 2$ after some experimentation because $f(1) = -6$, $f(2) = -1$, and $f(3) = 16$. Equation 3 becomes

$$x_{n+1} = x_n - \frac{x_n^3 - 2x_n - 5}{3x_n^2 - 2}$$

With $n = 1$ we have

$$\begin{aligned} x_2 &= x_1 - \frac{x_1^3 - 2x_1 - 5}{3x_1^2 - 2} \\ &= 2 - \frac{2^3 - 2(2) - 5}{3(2)^2 - 2} = 2.1 \end{aligned}$$

Then with $n = 2$ we obtain

$$x_3 = x_2 - \frac{x_2^3 - 2x_2 - 5}{3x_2^2 - 2} = 2.1 - \frac{(2.1)^3 - 2(2.1) - 5}{3(2.1)^2 - 2} \approx 2.0946$$

It turns out that this third approximation $x_3 \approx 2.0946$ is accurate to four decimal places. ■

Suppose that we want to achieve a given accuracy, say to eight decimal places, using Newton's method. How do we know when to stop? The rule of thumb that is generally used is that we can stop when successive approximations x_n and x_{n+1} agree to eight decimal places.

Notice that the procedure in going from n to $n + 1$ is the same for all values of n . (It is a recursive sequence, as defined in Section 1.6.) This means that Newton's method is particularly convenient for use with a programmable calculator or a computer.

EXAMPLE 6 | Find, correct to six decimal places, the root of the equation $\cos x = x$.

SOLUTION We first rewrite the equation in standard form:

$$\cos x - x = 0$$

Therefore we let $f(x) = \cos x - x$. Then $f'(x) = -\sin x - 1$, so Formula 3 becomes

$$x_{n+1} = x_n - \frac{\cos x_n - x_n}{-\sin x_n - 1} = x_n + \frac{\cos x_n - x_n}{\sin x_n + 1}$$

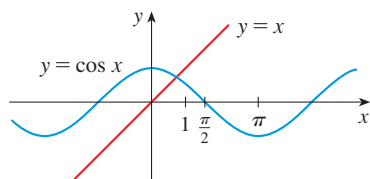


FIGURE 9

In order to guess a suitable value for x_1 we sketch the graphs of $y = \cos x$ and $y = x$ in Figure 9. It appears that they intersect at a point whose x -coordinate is somewhat less than 1, so let's take $x_1 = 1$ as a convenient first approximation. Then, remembering to put our calculator in radian mode, we get

$$x_2 \approx 0.75036387$$

$$x_3 \approx 0.73911289$$

$$x_4 \approx 0.73908513$$

$$x_5 \approx 0.73908513$$

Since x_4 and x_5 agree to six decimal places (eight, in fact), we conclude that the root of the equation, correct to six decimal places, is 0.739085. ■

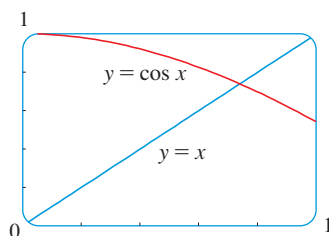


FIGURE 10

Instead of using the rough sketch in Figure 9 to get a starting approximation for Newton's method in Example 6, we could have used the more accurate graph that a calculator or computer provides. Figure 10 suggests that we use $x_1 = 0.75$ as the initial approximation. Then Newton's method gives

$$x_2 \approx 0.73911114 \quad x_3 \approx 0.73908513 \quad x_4 \approx 0.73908513$$

and so we obtain the same answer as before, but with one fewer step.

■ Taylor Polynomials

The tangent line approximation $L(x)$ is the best first-degree (linear) approximation to $f(x)$ near $x = a$ because $f(x)$ and $L(x)$ have the same rate of change (derivative) at a . For a better approximation than a linear one, let's try a second-degree (quadratic) approximation $P(x)$. In other words, we approximate a curve by a parabola instead of by a straight line. To make sure that the approximation is a good one, we stipulate the following:

- (i) $P(a) = f(a)$ (P and f should have the same value at a .)
- (ii) $P'(a) = f'(a)$ (P and f should have the same rate of change at a .)
- (iii) $P''(a) = f''(a)$ (The slopes of P and f should change at the same rate at a .)

Let's write the polynomial P in the form

$$P(x) = A + B(x - a) + C(x - a)^2$$

Then

$$P'(x) = B + 2C(x - a) \quad \text{and} \quad P''(x) = 2C$$

Applying the conditions (i), (ii), and (iii), we get

$$P(a) = f(a) \quad \Rightarrow \quad A = f(a)$$

$$P'(a) = f'(a) \quad \Rightarrow \quad B = f'(a)$$

$$P''(a) = f''(a) \quad \Rightarrow \quad 2C = f''(a) \quad \Rightarrow \quad C = \frac{1}{2}f''(a)$$

So the quadratic function that satisfies the three conditions is

(4)

$$P(x) = f(a) + f'(a)(x - a) + \frac{1}{2}f''(a)(x - a)^2$$

This function is called the **second-degree Taylor polynomial of f centered at a** and is usually denoted by $T_2(x)$.

EXAMPLE 7 | Find the second-degree Taylor polynomial $T_2(x)$ centered at $a = 0$ for the function $f(x) = \cos x$. Illustrate by graphing T_2 , f , and the linearization $L(x) = 1$.

SOLUTION Since $f(x) = \cos x$, $f'(x) = -\sin x$, and $f''(x) = -\cos x$, the second-degree Taylor polynomial centered at 0 is

$$\begin{aligned} T_2(x) &= f(0) + f'(0)x + \frac{1}{2}f''(0)x^2 \\ &= 1 + 0 + \frac{1}{2}(-1)x^2 = 1 - \frac{1}{2}x^2 \end{aligned}$$

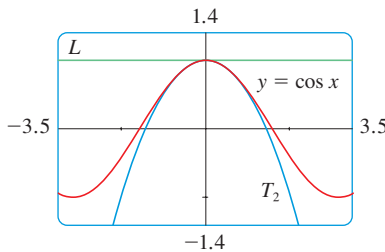


FIGURE 11

Figure 11 shows a graph of the cosine function together with its linear approximation $L(x) = 1$ and its quadratic approximation $T_2(x) = 1 - \frac{1}{2}x^2$ near 0. You can see that the quadratic approximation is much better than the linear one. ■

Instead of being satisfied with a linear or quadratic approximation to $f(x)$ near a , let's try to find better approximations with higher-degree polynomials. We look for an n th-degree polynomial

$$(5) \quad T_n(x) = c_0 + c_1(x - a) + c_2(x - a)^2 + c_3(x - a)^3 + \cdots + c_n(x - a)^n$$

such that T_n and its first n derivatives have the same values at $x = a$ as f and its first n derivatives. By differentiating repeatedly and setting $x = a$, you are asked to show in Exercise 44 that these conditions are satisfied if $c_0 = f(a)$, $c_1 = f'(a)$, $c_2 = \frac{1}{2}f''(a)$, and in general

$$c_k = \frac{f^{(k)}(a)}{k!}$$

where $k! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot \cdots \cdot k$ is called k factorial. The resulting polynomial

$$T_n(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n$$

is called the **n th-degree Taylor polynomial of f centered at a** .

EXAMPLE 8 | Find the first three Taylor polynomials T_1 , T_2 , and T_3 for the function $f(x) = \ln x$ centered at $a = 1$.

SOLUTION We start by calculating the first three derivatives at $a = 1$:

$$\begin{array}{llll} f(x) = \ln x & f'(x) = \frac{1}{x} & f''(x) = -\frac{1}{x^2} & f'''(x) = \frac{2}{x^3} \\ f(1) = 0 & f'(1) = 1 & f''(1) = -1 & f'''(1) = 2 \end{array}$$

Then

$$T_1(x) = f(1) + f'(1)(x - 1) = x - 1$$

$$T_2(x) = T_1(x) + \frac{f''(1)}{2!}(x - 1)^2 = x - 1 - \frac{1}{2}(x - 1)^2$$

$$T_3(x) = T_2(x) + \frac{f'''(1)}{3!}(x - 1)^3 = x - 1 - \frac{1}{2}(x - 1)^2 + \frac{1}{3}(x - 1)^3$$

Figure 12 shows the graphs of these Taylor polynomials. Notice that these polynomial approximations are better when x is close to 1 and that each successive approximation is better than the preceding ones.

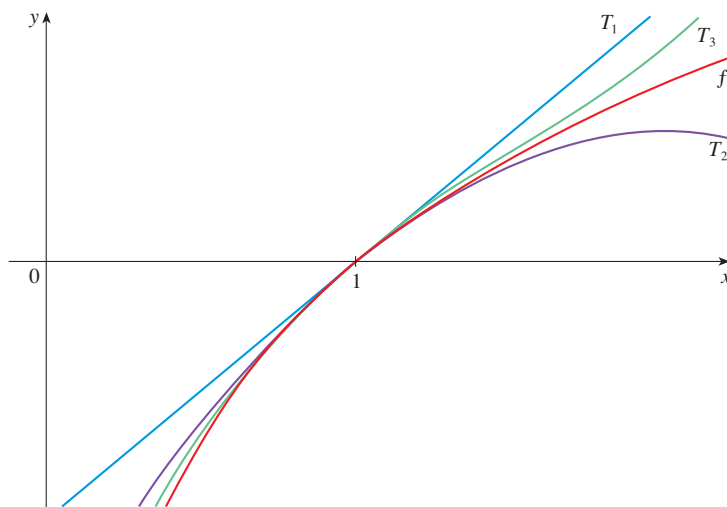


FIGURE 12

EXERCISES 3.8

1–4 Find the linearization $L(x)$ of the function at a .

1. $f(x) = x^4 + 3x^2$, $a = -1$ 2. $f(x) = \ln x$, $a = 1$
 3. $f(x) = \cos x$, $a = \pi/2$ 4. $f(x) = x^{3/4}$, $a = 16$

5. Find the linear approximation of the function $f(x) = \sqrt{1-x}$ at $a = 0$ and use it to approximate the numbers $\sqrt{0.9}$ and $\sqrt{0.99}$. Illustrate by graphing f and the tangent line.

6. Find the linear approximation of the function $g(x) = \sqrt[3]{1+x}$ at $a = 0$ and use it to approximate the numbers $\sqrt[3]{0.95}$ and $\sqrt[3]{1.1}$. Illustrate by graphing g and the tangent line.

7–10 Verify the given linear approximation at $a = 0$. Then determine the values of x for which the linear approximation is accurate to within 0.1.

7. $\sqrt[3]{1-x} \approx 1 - \frac{1}{3}x$

8. $\tan x \approx x$

9. $1/(1+2x)^4 \approx 1 - 8x$

10. $e^x \approx 1 + x$

11–12 Use a linear approximation to estimate the given number.

11. $(2.001)^5$

12. $e^{-0.015}$

13–14 Explain, in terms of linear approximations, why the approximation is reasonable.

13. $\ln 1.05 \approx 0.05$

14. $(1.01)^6 \approx 1.06$

15. Insecticide resistance If the frequency of a gene for insecticide resistance is p (a constant), then its frequency in the next generation is given by the expression

$$f = \frac{p(1+s)}{1+sp}$$

where s is the reproductive advantage this gene has over the

wild type in the presence of the insecticide. Often the selective advantage s is very small. Approximate the frequency in the next generation with a linear approximation, given that s is small.

- 16. Relative change in blood velocity** Suppose $y = f(x)$ and x and y change by amounts Δx and Δy . A way of expressing a linear approximation is to write $\Delta y \approx f'(x) \Delta x$. The *relative change* in y is $\Delta y/y$.

A special case of Poiseuille's law of laminar flow (see Example 3.3.9) is that at the central axis of a blood vessel the velocity of the blood is related to the radius R of the vessel by an equation of the form $v = cR^2$. If the radius changes, how is the relative change in the blood velocity related to the relative change in the radius? If the radius is increased by 10%, what happens to the velocity?

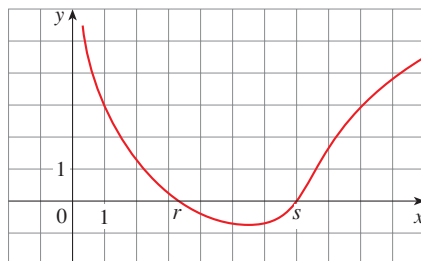
- 17. Relative change in blood flow** Another law of Poiseuille says that when blood flows along a blood vessel, the flux F (the volume of blood per unit time that flows past a given point) is proportional to the fourth power of the radius R of the blood vessel:

$$F = kR^4$$

(We will show why this is true in Section 6.3.) A partially clogged artery can be expanded by an operation called angioplasty, in which a balloon-tipped catheter is inflated inside the artery in order to widen it and restore the normal blood flow.

Show that the relative change in F is about four times the relative change in R . How will a 5% increase in the radius affect the flow of blood?

- 18. Volume and surface area of a tumor** The diameter of a tumor was measured to be 19 mm. If the diameter increases by 1 mm, use linear approximations to estimate the relative changes in the volume ($V = \frac{4}{3}\pi r^3$) and surface area ($S = 4\pi r^2$).
- 19.** The figure shows the graph of a function f . Suppose that Newton's method is used to approximate the root r of the equation $f(x) = 0$ with initial approximation $x_1 = 1$.
- Draw the tangent lines that are used to find x_2 and x_3 , and estimate the numerical values of x_2 and x_3 .
 - Would $x_1 = 5$ be a better first approximation? Explain.



- 20.** Follow the instructions for Exercise 19(a) but use $x_1 = 9$ as the starting approximation for finding the root s .

21–22 Use Newton's method with the specified initial approximation x_1 to find x_3 , the third approximation to the root of the given equation. (Give your answer to four decimal places.)

21. $x^3 + 2x - 4 = 0$, $x_1 = 1$

22. $\frac{1}{3}x^3 + \frac{1}{2}x^2 + 3 = 0$, $x_1 = -3$

23–26 Use Newton's method to find all roots of the equation correct to six decimal places.

23. $x^4 = 1 + x$

24. $e^x = 3 - 2x$

25. $(x - 2)^2 = \ln x$

26. $\frac{1}{x} = 1 + x^3$

 **27–31** Use Newton's method to find all the roots of the equation correct to eight decimal places. Start by drawing a graph to find initial approximations.

27. $x^6 - x^5 - 6x^4 - x^2 + x + 10 = 0$

28. $x^2(4 - x^2) = \frac{4}{x^2 + 1}$

29. $x^2\sqrt{2 - x - x^2} = 1$

30. $3 \sin(x^2) = 2x$

31. $4e^{-x^2} \sin x = x^2 - x + 1$

32. Infectious disease outbreak size If 99% of a population is initially uninfected and each initial infected person generates, on average, two new infections, then, according to the model we considered in Example 3.5.13,

$$0.99e^{-2A} = 1 - A$$

where A is the fraction of the population infected at the end of an outbreak. Use Newton's method to obtain an approximation (accurate to two decimal places) for the percentage of the population that is eventually infected.

33. Explain why Newton's method doesn't work for finding the root of the equation $x^3 - 3x + 6 = 0$ if the initial approximation is chosen to be $x_1 = 1$.

- 34.** (a) Use Newton's method with $x_1 = 1$ to find the root of the equation $x^3 - x = 1$ correct to six decimal places.
 (b) Solve the equation in part (a) using $x_1 = 0.6$ as the initial approximation.
 (c) Solve the equation in part (a) using $x_1 = 0.57$. (You definitely need a programmable calculator for this part.)
 (d) Graph $f(x) = x^3 - x - 1$ and its tangent lines at $x_1 = 1$, 0.6 , and 0.57 to explain why Newton's method is so sensitive to the value of the initial approximation.

35–38 Find the Taylor polynomial of degree n centered at the number a .

35. $f(x) = e^x$, $n = 3$, $a = 0$

36. $f(x) = \sin \pi x$, $n = 3$, $a = 0$

37. $f(x) = 1/x$, $n = 4$, $a = 1$

38. $f(x) = \sqrt{x}$, $n = 2$, $a = 4$

-  39. Find the quadratic approximation to $f(x) = \sqrt{x+3}$ near $a = 1$. Graph f , the quadratic approximation, and the linear approximation from Example 1 on a common screen. What do you conclude?
-  40. Determine the values of x for which the quadratic approximation $f(x) \approx T_2(x)$ in Example 7 is accurate to within 0.1. [Hint: Graph $y = T_2(x)$, $y = \cos x - 0.1$, and $y = \cos x + 0.1$ on a common screen.]
-  41. Find the first five Taylor polynomials for $f(x) = \sin x$ centered at 0. Graph them on the interval $[-4, 4]$ and comment on how well they approximate f .
-  42. Find the 8th-degree Taylor polynomial centered at $a = 0$ for the function $f(x) = \cos x$. Graph f together with the Taylor polynomials T_2, T_4, T_6, T_8 in the viewing rectangle $[-5, 5]$

by $[-1.4, 1.4]$ and comment on how well they approximate f .

43. Habitat fragmentation and species conservation

The size of a class-structured population is modeled in Section 8.8. In certain situations the long-term per capita growth rate of the population is given by

$$r = \frac{1}{2}(1 + \sqrt{1 + 8s})$$

where s is the annual survival probability of juveniles.

- Approximate the growth rate with a Taylor polynomial of degree one (linear approximation) centered at 0.
- Approximate the growth rate with a Taylor polynomial of degree two centered at 0.

44. Show that if a polynomial T_n of the form given in Equation 5 has the same value at a and the same derivatives at $x = a$ as a function f , then its coefficients are given by the formula

$$c_k = \frac{f^{(k)}(a)}{k!}$$

PROJECT Harvesting Renewable Resources



In Exercise 1.6.32 we considered the Ricker difference equation

$$x_{t+1} = cx_t e^{-x_t}$$

where x_t is the size of a population at time t and c is the per capita reproductive output when the population size is small. (We assume that $c > 1$, which means that individuals more than replace themselves when the population size is small.)

- Suppose that as $t \rightarrow \infty$ the population size approaches a limiting value x . Express x in terms of the growth factor c . You will find two solutions; focus only on the one that is strictly positive.
- Now let's consider harvesting. Assume that h individuals are harvested in each time step. Then our model becomes

$$x_{t+1} = cx_t e^{-x_t} - h$$

What equation does the limiting population size x satisfy?

- Even though the equation you found in Problem 2 can't be solved explicitly for x , you can use implicit differentiation to find an expression for the derivative of x with respect to h . Do so.
- Our aim is to find an expression for the limiting population size x in terms of the harvest rate h . As a first approximation, find the linearization of the function $x(h)$ at $h = 0$. [Note that $x(0)$ is known from Problem 1.]
- Find the second-order Taylor polynomial approximation for $x(h)$ at $h = 0$.

Chapter 3 REVIEW

CONCEPT CHECK

- Write an expression for the slope of the tangent line to the curve $y = f(x)$ at the point $(a, f(a))$.
- Define the derivative $f'(a)$. Discuss two ways of interpreting this number.
- If $y = f(x)$ and x changes from x_1 to x_2 , write expressions for the following.
 - The average rate of change of y with respect to x over the interval $[x_1, x_2]$
 - The instantaneous rate of change of y with respect to x at $x = x_1$
- Define the second derivative of f . If $f(t)$ is the position function of a particle, how can you interpret the second derivative?
- What does it mean for f to be differentiable at a ?
 - What is the relation between the differentiability and continuity of a function?
 - Sketch the graph of a function that is continuous but not differentiable at $a = 2$.
- Describe several ways in which a function can fail to be differentiable. Illustrate with sketches.
- State each differentiation rule both in symbols and in words.

(a) The Power Rule	(b) The Constant Multiple Rule
(c) The Sum Rule	(d) The Difference Rule
- | | |
|----------------------|-----------------------|
| (e) The Product Rule | (f) The Quotient Rule |
| (g) The Chain Rule | |
- State the derivative of each function.

(a) $y = x^n$	(b) $y = e^x$	(c) $y = b^x$
(d) $y = \ln x$	(e) $y = \log_b x$	(f) $y = \sin x$
(g) $y = \cos x$	(h) $y = \tan x$	(i) $y = \csc x$
(j) $y = \sec x$	(k) $y = \cot x$	(l) $y = \tan^{-1} x$
- How is the number e defined?
 - Express e as a limit.
 - Why is the natural exponential function $y = e^x$ used more often in calculus than the other exponential functions $y = b^x$?
 - Why is the natural logarithmic function $y = \ln x$ used more often in calculus than the other logarithmic functions $y = \log_b x$?
- Explain how implicit differentiation works. When should you use it?
 - Explain how logarithmic differentiation works. When should you use it?
- Write an expression for the linearization of f at a .
- Write an expression for the n th-degree Taylor polynomial of f centered at a .

Answers to the Concept Check can be found on the back endpapers.

TRUE-FALSE QUIZ

Determine whether the statement is true or false. If it is true, explain why. If it is false, explain why or give an example that disproves the statement.

- If f is continuous at a , then f is differentiable at a .
- If $f'(r)$ exists, then $\lim_{x \rightarrow r} f(x) = f(r)$.
- If f and g are differentiable, then

$$\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$$

- If f and g are differentiable, then

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g'(x)$$

- If f and g are differentiable, then

$$\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x)$$

$$6. \text{ If } f \text{ is differentiable, then } \frac{d}{dx} \sqrt{f(x)} = \frac{f'(x)}{2\sqrt{f(x)}}.$$

$$7. \text{ If } f \text{ is differentiable, then } \frac{d}{dx} f(\sqrt{x}) = \frac{f'(x)}{2\sqrt{x}}.$$

$$8. \text{ If } y = e^2, \text{ then } y' = 2e.$$

$$9. \frac{d}{dx}(10^x) = x10^{x-1} \qquad 10. \frac{d}{dx}(\ln 10) = \frac{1}{10}$$

$$11. \frac{d}{dx}(\tan^2 x) = \frac{d}{dx}(\sec^2 x) \qquad 12. \frac{d^2 y}{dx^2} = \left(\frac{dy}{dx}\right)^2$$

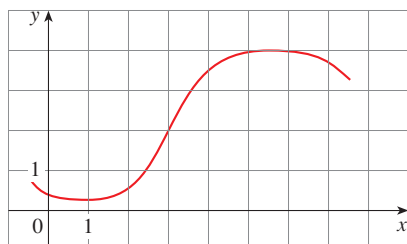
$$13. \text{ If } g(x) = x^5, \text{ then } \lim_{x \rightarrow 2} \frac{g(x) - g(2)}{x - 2} = 80$$

$$14. \text{ An equation of the tangent line to the parabola } y = x^2 \text{ at } (-2, 4) \text{ is } y - 4 = 2x(x + 2).$$

EXERCISES

1. For the function f whose graph is shown, arrange the following numbers in increasing order:

$$0 \quad 1 \quad f'(2) \quad f'(3) \quad f'(5) \quad f''(5)$$

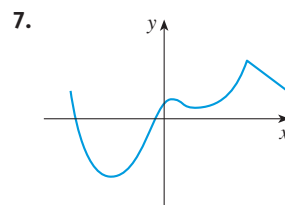
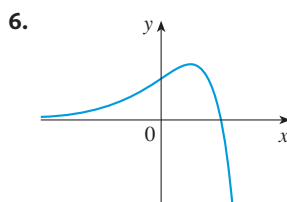
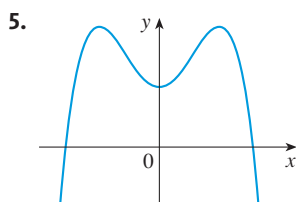


2. **Life expectancy** The table shows how the life expectancy $L(t)$ in Bangladesh has changed from 1990 to 2010.

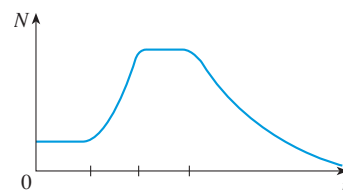
t	1990	1995	2000	2005	2010
$L(t)$	56	61	65	68	71

- (a) Calculate the average rate of change of the life expectancy $L(t)$ with respect to time over the following time intervals.
- (i) [1990, 2000] (ii) [1995, 2000]
 - (iii) [2000, 2010] (iv) [2000, 2005]
- (b) Estimate the value of $L'(2000)$.
3. The total cost of repaying a student loan at an interest rate of $r\%$ per year is $C = f(r)$.
- (a) What is the meaning of the derivative $f'(r)$? What are its units?
 - (b) What does the statement $f'(10) = 1200$ mean?
 - (c) Is $f'(r)$ always positive or does it change sign?
4. (a) Use the definition of a derivative to find $f'(2)$, where $f(x) = x^3 - 2x$.
- (b) Find an equation of the tangent line to the curve $y = x^3 - 2x$ at the point $(2, 4)$.
- (c) Illustrate part (b) by graphing the curve and the tangent line on the same screen.

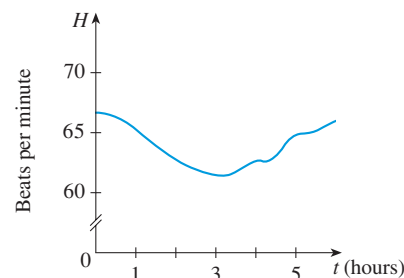
5–7 Trace or copy the graph of the function. Then sketch a graph of its derivative directly beneath.



8. **Bacteria count** Shown is a typical graph of the number N of bacteria grown in a bacteria culture as a function of time t .
- (a) What is the meaning of the derivative $N'(t)$?
 - (b) Sketch the graph of $N'(t)$.

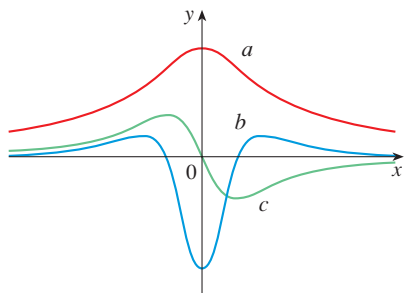


9. **Antihypertension medication** The figure shows the heart rate $H(t)$ after a patient has taken nifedipine tablets.
- (a) What is the meaning of the derivative $H'(t)$?
 - (b) Sketch the graph of $H'(t)$.

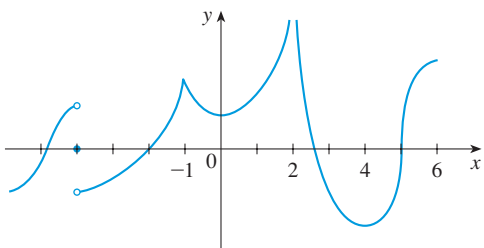


10. (a) Find the asymptotes of the graph of $f(x) = \frac{4-x}{3+x}$ and use them to sketch the graph.
- (b) Use your graph from part (a) to sketch the graph of f' .
- (c) Use the definition of a derivative to find $f'(x)$.
- (d) Use a calculator to graph f' and compare with your sketch in part (b).
11. (a) If $f(x) = \sqrt{3-5x}$, use the definition of a derivative to find $f'(x)$.
- (b) Find the domains of f and f' .
- (c) Graph f and f' on a common screen. Compare the graphs to see whether your answer to part (a) is reasonable.

-  12. The figure shows the graphs of f , f' , and f'' . Identify each curve, and explain your choices.

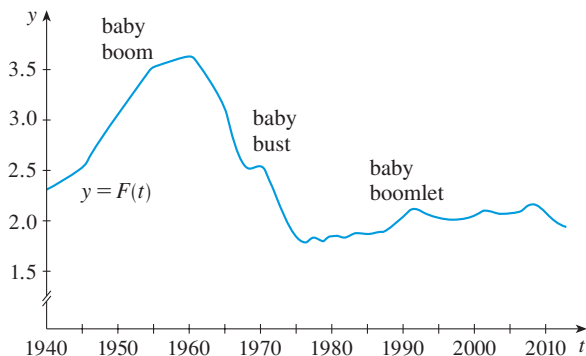


13. The graph of f is shown. State, with reasons, the numbers at which f is not differentiable.



14. The **total fertility rate** at time t , denoted by $F(t)$, is an estimate of the average number of children born to each woman (assuming that current birth rates remain constant). The graph of the total fertility rate in the United States shows the fluctuations from 1940 to 2010.

- (a) Estimate the values of $F'(1950)$, $F'(1965)$, and $F'(1987)$.
 (b) What are the meanings of these derivatives?
 (c) Can you suggest reasons for the values of these derivatives?



15–50 Calculate y' .

15. $y = (x^4 - 3x^2 + 5)^3$

16. $y = \cos(\tan x)$

17. $y = \sqrt{x} + \frac{1}{\sqrt[3]{x^4}}$

18. $y = \frac{3x - 2}{\sqrt{2x + 1}}$

19. $y = 2x\sqrt{x^2 + 1}$

20. $y = \frac{e^x}{1 + x^2}$

21. $y = e^{\sin 2\theta}$

22. $y = e^{-t}(t^2 - 2t + 2)$

23. $y = \frac{t}{1 - t^2}$

24. $y = e^{mx} \cos nx$

25. $y = \frac{e^{1/x}}{x^2}$

26. $y = \left(\frac{u - 1}{u^2 + u + 1} \right)^4$

27. $xy^4 + x^2y = x + 3y$

28. $y = \ln(\csc 5x)$

29. $y = \frac{\sec 2\theta}{1 + \tan 2\theta}$

30. $x^2 \cos y + \sin 2y = xy$

31. $y = e^{cx}(c \sin x - \cos x)$

32. $y = \ln(x^2 e^x)$

33. $y = \log_5(1 + 2x)$

34. $y = (\ln x)^{\cos x}$

35. $\sin(xy) = x^2 - y$

36. $y = \sqrt{t \ln(t^4)}$

37. $y = 3^{x \ln x}$

38. $xe^y = y - 1$

39. $y = \ln \sin x - \frac{1}{2} \sin^2 x$

40. $y = \frac{(x^2 + 1)^4}{(2x + 1)^3(3x - 1)^5}$

41. $y = x \tan^{-1}(4x)$

42. $y = e^{\cos x} + \cos(e^x)$

43. $y = \ln |\sec 5x + \tan 5x|$

44. $y = 10^{\tan \pi \theta}$

45. $y = \tan^2(\sin \theta)$

46. $y = \ln \left| \frac{x^2 - 4}{2x + 5} \right|$

47. $y = \sin(\tan \sqrt{1 + x^3})$

48. $y = \arctan(\arcsin \sqrt{x})$

49. $y = \cos(e^{\sqrt{\tan 3x}})$

50. $y = \sin^2(\cos \sqrt{\sin \pi x})$

51. If $f(t) = \sqrt{4t + 1}$, find $f''(2)$.

52. If $g(\theta) = \theta \sin \theta$, find $g''(\pi/6)$.

53. If $f(x) = 2^x$, find $f^{(n)}(x)$.

54. Find $f^{(n)}(x)$ if $f(x) = 1/(2 - x)$.

55–56 Find an equation of the tangent to the curve at the given point.

55. $y = 4 \sin^2 x$, $(\pi/6, 1)$

56. $y = \frac{x^2 - 1}{x^2 + 1}$, $(0, -1)$

57–58 Find equations of the tangent line and normal line to the curve at the given point.

57. $y = (2 + x)e^{-x}$, $(0, 2)$

58. $x^2 + 4xy + y^2 = 13$, $(2, 1)$

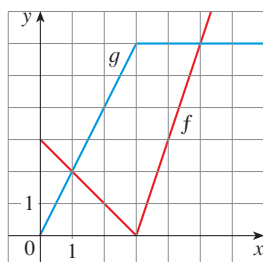
59. (a) If $f(x) = x\sqrt{5 - x}$, find $f'(x)$.

(b) Find equations of the tangent lines to the curve $y = x\sqrt{5 - x}$ at the points $(1, 2)$ and $(4, 4)$.

-  (c) Illustrate part (b) by graphing the curve and tangent lines on the same screen.
 (d) Check to see that your answer to part (a) is reasonable by comparing the graphs of f and f' .
 60. (a) Graph the function $f(x) = x - 2 \sin x$ in the viewing rectangle $[0, 8]$ by $[-2, 8]$.
 (b) On which interval is the average rate of change larger: $[1, 2]$ or $[2, 3]$?
 (c) At which value of x is the instantaneous rate of change larger: $x = 2$ or $x = 5$?
 (d) Check your visual estimates in part (c) by computing $f'(x)$ and comparing the numerical values of $f'(2)$ and $f'(5)$.

61. Suppose that $h(x) = f(x)g(x)$ and $F(x) = f(g(x))$, where $f(2) = 3$, $g(2) = 5$, $g'(2) = 4$, $f'(2) = -2$, and $f'(5) = 11$. Find (a) $h'(2)$ and (b) $F'(2)$.

62. If f and g are the functions whose graphs are shown, let $P(x) = f(x)g(x)$, $Q(x) = f(x)/g(x)$, and $C(x) = f(g(x))$. Find (a) $P'(2)$, (b) $Q'(2)$, and (c) $C'(2)$.



63–70 Find f' in terms of g' .

- | | |
|-------------------------|-----------------------|
| 63. $f(x) = x^2g(x)$ | 64. $f(x) = g(x^2)$ |
| 65. $f(x) = [g(x)]^2$ | 66. $f(x) = g(g(x))$ |
| 67. $f(x) = g(e^x)$ | 68. $f(x) = e^{g(x)}$ |
| 69. $f(x) = \ln g(x) $ | 70. $f(x) = g(\ln x)$ |

71. At what point on the curve $y = [\ln(x + 4)]^2$ is the tangent horizontal?
72. (a) Find an equation of the tangent to the curve $y = e^x$ that is parallel to the line $x - 4y = 1$.
 (b) Find an equation of the tangent to the curve $y = e^x$ that passes through the origin.
73. Find the points on the ellipse $x^2 + 2y^2 = 1$ where the tangent line has slope 1.
74. Find a parabola $y = ax^2 + bx + c$ that passes through the point $(1, 4)$ and whose tangent lines at $x = -1$ and $x = 5$ have slopes 6 and -2 , respectively.
75. An equation of motion of the form $s = Ae^{-ct} \cos(\omega t + \delta)$ represents damped oscillation of an object. Find the velocity and acceleration of the object.

76. A particle moves along a horizontal line so that its coordinate at time t is $x = \sqrt{b^2 + c^2 t^2}$, $t \geq 0$, where b and c are positive constants.

- (a) Find the velocity and acceleration functions.
 (b) Show that the particle always moves in the positive direction.

77. The volume of a right circular cone is $V = \frac{1}{3}\pi r^2 h$, where r is the radius of the base and h is the height.

- (a) Find the rate of change of the volume with respect to the height if the radius is constant.
 (b) Find the rate of change of the volume with respect to the radius if the height is constant.

78. The **Michaelis-Menten equation** for the enzyme pepsin is

$$v = \frac{0.50[S]}{3.0 \times 10^{-4} + [S]}$$

where v is the rate of an enzymatic reaction and $[S]$ is the concentration of a substrate S . Calculate $dv/d[S]$ and interpret it.

79. **Health care expenditures** The US health care expenditures for 1970–2008 have been modeled by the function

$$E(t) = 101.35e^{0.088128t}$$

where t is the number of years elapsed since 1970 and E is measured in billions of dollars. According to this model, at what rate were health care expenditures increasing in 1980? In 2000?

80. **Drug concentration** The function $C(t) = K(e^{-at} - e^{-bt})$, where a , b , and K are positive constants and $b > a$, is used to model the concentration at time t of a drug injected into the bloodstream.

- (a) Show that $\lim_{t \rightarrow \infty} C(t) = 0$.
 (b) Find $C'(t)$, the rate of change of drug concentration in the blood.
 (c) When is this rate equal to 0?

81. **Bacteria growth** A bacteria culture contains 200 cells initially and grows at a rate proportional to its size. After half an hour the population has increased to 360 cells.

- (a) Find the number of bacteria after t hours.
 (b) Find the number of bacteria after 4 hours.
 (c) Find the rate of growth after 4 hours.
 (d) When will the population reach 10,000?

82. Cobalt-60 has a half-life of 5.24 years.

- (a) Find the mass that remains from a 100-mg sample after 20 years.
 (b) How long would it take for the mass to decay to 1 mg?

83. **Drug elimination** Let $C(t)$ be the concentration of a drug in the bloodstream. As the body eliminates the drug, $C(t)$ decreases at a rate that is proportional to the amount of the drug that is present at the time. Thus $C'(t) = -kC(t)$,

where k is a positive number called the *elimination constant* of the drug.

- If C_0 is the concentration at time $t = 0$, find the concentration at time t .
- If the body eliminates half the drug in 30 hours, how long does it take to eliminate 90% of the drug?

- 84.** A cup of hot chocolate has temperature 80°C in a room kept at 20°C . After half an hour the hot chocolate cools to 60°C .
- What is the temperature of the chocolate after another half hour?
 - When will the chocolate have cooled to 40°C ?
- 85.** The volume of a cube is increasing at a rate of $10\text{ cm}^3/\text{min}$. How fast is the surface area increasing when the length of an edge is 30 cm ?

- 86. Yeast population** The number of yeast cells in a laboratory culture increases rapidly initially but levels off eventually. The population is modeled by the function

$$n = f(t) = \frac{a}{1 + be^{-0.7t}}$$

where t is measured in hours. At time $t = 0$ the population is 20 cells and is increasing at a rate of 12 cells/hour. Find the values of a and b . According to this model, what happens to the yeast population in the long run?

- 87.** Use Newton's method to find the root of the equation $x^5 - x^4 + 3x^2 - 3x - 2 = 0$ in the interval $[1, 2]$ correct to six decimal places.
- 88.** Use Newton's method to find all roots of the equation $\sin x = x^2 - 3x + 1$ correct to six decimal places.
- 89.** (a) Find the linearization of $f(x) = \sqrt[3]{1 + 3x}$ at $a = 0$. State the corresponding linear approximation and use it to give an approximate value for $\sqrt[3]{1.03}$.
 (b) Determine the values of x for which the linear approximation given in part (a) is accurate to within 0.1.
- 90.** (a) Find the first three Taylor polynomials for $f(x) = 4(x - 2)^{-2}$ centered at 0.
 (b) Graph f and the Taylor polynomials from part (a) on the interval $[-1, 1]$ and comment on how well the polynomials approximate f .

- 91. Dialysis** The project on page 458 models the removal of urea from the bloodstream via dialysis. In certain situations the duration of dialysis required, given that the initial urea concentration is c , where $c > 1$, is given by the equation

$$t = \ln\left(\frac{3c + \sqrt{9c^2 - 8c}}{2}\right)$$

- Use a linear approximation to estimate the time required if the initial concentration is near $c = 1$.
- Use a second-order Taylor polynomial to give a more accurate approximation.

- 92. Infectious disease outbreak size** We have worked with the model

$$\rho e^{-qA} = 1 - A$$

where A is the fraction of the population infected, q is a measure of disease transmissibility, and ρ is the fraction of the population that is initially susceptible to infection.

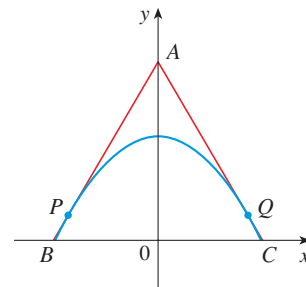
- Use implicit differentiation to find the linear approximation of A as a function of q at $q = 0$.
- Find the second-order Taylor polynomial approximation for $A(q)$ at $q = 0$.

- 93.** Express the limit

$$\lim_{\theta \rightarrow \pi/3} \frac{\cos \theta - 0.5}{\theta - \pi/3}$$

as a derivative and thus evaluate it.

- 94.** Find points P and Q on the parabola $y = 1 - x^2$ so that the triangle ABC formed by the x -axis and the tangent lines at P and Q is an equilateral triangle.

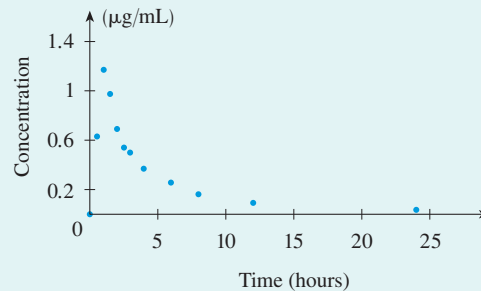


CASE STUDY 1b Kill Curves and Antibiotic Effectiveness



We are studying the relationship between the magnitude of antibiotic treatment and the effectiveness of the treatment. Recall that the extent of bacterial killing by an antibiotic is determined by both the *antibiotic concentration profile* and the *dose response relationship*. Figure 1 shows the antibiotic concentration profile for ciprofloxacin.¹ Our first goal here is to choose an appropriate mathematical description of this profile.

FIGURE 1
Antibiotic concentration profile in plasma of a healthy human volunteer after receiving 500 mg of ciprofloxacin



In Case Study 1a we modeled the initial increase in concentration as occurring instantly. We then need to determine how to model the decay in concentration. From Figure 1 it looks as though the rate of decay (that is, the slope of the relationship between concentration and time) is smaller for lower concentrations. We also know that the rate of decay must be zero when the concentration is zero. Therefore, as a simple model, let's suppose that the rate of decay of concentration is proportional to the current concentration; that is,

$$(1) \quad \frac{dc}{dt} = -kc$$

for some positive constant k . Here c is measured in $\mu\text{g/mL}$ and t is measured in hours.

1. If the concentration at $t = 0$ is c_0 , verify that the concentration function $c(t) = c_0 e^{-kt}$ satisfies Equation 1. Suppose that the antibiotic ciprofloxacin has a half-life of 4 hours. What is the value of k ?

Next we wish to model the bacteria population dynamics. When a bacteria population is small, it grows at a rate that is proportional to its size because each bacterium produces a constant number of offspring per unit time. A simple model for the growth of the bacteria population size P when small is therefore

$$(2) \quad \frac{dP}{dt} = rP$$

where r is a constant called the *per capita growth rate* (it is the rate of offspring production by each individual bacterium). As a result, if the bacteria population starts at size P_0 , its predicted size at time t is $P(t) = P_0 e^{rt}$.

1. Adapted from Imre, S. et al., "Validation of an HPLC Method for the Determination of Ciprofloxacin in Human Plasma," *Journal of Pharmaceutical and Biomedical Analysis* 33 (2003): 125–30.

As the population grows, resources become depleted. Eventually the bacteria population reaches a size at which it no longer changes. For the data in Figure 2² it looks as though the maximum population size is around 12 CFU/mL. A simple model is therefore that the population grows according to Equation 2 if $P < 12$ and it remains constant at $P = 12$ if the value of P predicted from the model in Equation 2 is ever greater than or equal to 12.

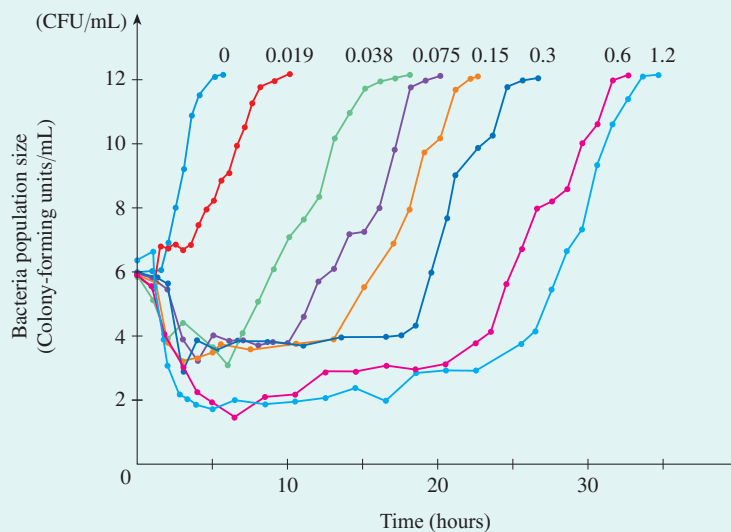


FIGURE 2

The kill curves of ciprofloxacin for *E. coli* when measured in a growth chamber. The concentration of ciprofloxacin at $t = 0$ is indicated above each curve (in $\mu\text{g/mL}$).

Our final step is to connect the model for bacteria population growth to the model for the antibiotic concentration profile. The connection between the two is given by the dose response relationship. Recall that in Case Study 1a we modeled the dose response relationship with the piecewise defined function

$$r(c) = \begin{cases} r_2 & \text{if } c < MIC \\ r_1 & \text{if } c \geq MIC \end{cases}$$

where $r(c)$ is the per capita growth rate of the bacteria population and MIC is a constant referred to as the *minimum inhibitory concentration* ($MIC = 0.013 \mu\text{g/mL}$ in this case). The constants r_1 and r_2 give the per capita growth rate under high and low antibiotic concentrations, respectively, with $r_1 < 0$ and $r_2 > 0$ (Figure 3).

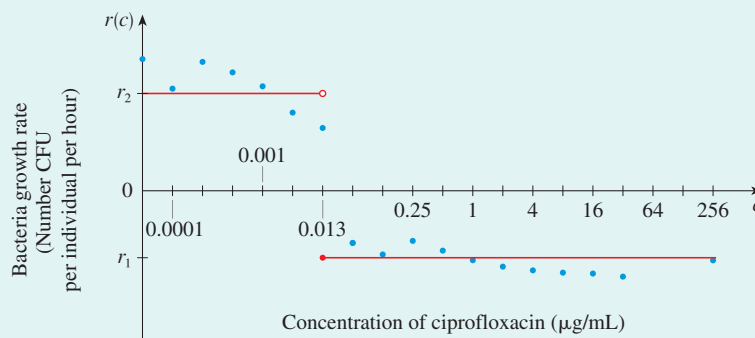


FIGURE 3

Dose response relationship modeled by the piecewise defined function $r(c)$

Source: Adapted from W. Bär et al., “Rapid Method for Detection of Minimal Bactericidal Concentration of Antibiotics,” *Journal of Microbiological Methods* 77 (2009): 85–89, Figure 1.

2. Adapted from A. Firsov et al., “Parameters of Bacterial Killing and Regrowth Kinetics and Antimicrobial Effect Examined in Terms of Area Under the Concentration–Time Curve Relationships: Action of Ciprofloxacin against *Escherichia coli* in an In Vitro Dynamic Model,” *Antimicrobial Agents and Chemotherapy* 41 (1997): 1281–87.

Suppose the bacteria population starts at $t = 0$ at a size of 6 CFU/mL. Suppose also that $MIC = 0.013$, $k = 0.175$, $r_1 = -\frac{1}{20}$, and $r_2 = \frac{1}{3}$.

2. Using the form of the solution to Equation 2, show that the bacteria population size at time t is given by the function

$$(3a) \quad P(t) = \begin{cases} 6e^{t/3} & \text{if } t < 2.08 \\ 12 & \text{if } t \geq 2.08 \end{cases}$$

if $c_0 < MIC$, where $MIC = 0.013$. On the other hand, if $c_0 > MIC$, show that

$$(3b) \quad P(t) = \begin{cases} 6e^{-t/20} & \text{if } t < a \\ 6Ae^{t/3} & \text{if } a \leq t < b \\ 12 & \text{if } t \geq b \end{cases}$$

where the constants a , b , and A are defined by $a = 5.7 \ln(77c_0)$, $b = 6.6 \ln(77c_0) + 2.08$, and $A = (77c_0)^{-2.2}$.

Equations 3 are the predicted kill curves explored in Case Study 1a.

3. Using the form of the solution to Equation 2, show that for arbitrary MIC , k , r_1 , and r_2 the bacteria population size at time t is given by

$$(4a) \quad P(t) = \begin{cases} 6e^{r_2 t} & \text{if } t < t_2 \\ 12 & \text{if } t \geq t_2 \end{cases}$$

if $c_0 < MIC$, and

$$(4b) \quad P(t) = \begin{cases} 6e^{r_1 t} & \text{if } t < t_1 \\ 6e^{r_1 t_1} e^{r_2(t-t_1)} & \text{if } t_1 \leq t < \left(1 - \frac{r_1}{r_2}\right)t_1 + t_2 \\ 12 & \text{if } \left(1 - \frac{r_1}{r_2}\right)t_1 + t_2 \leq t \end{cases}$$

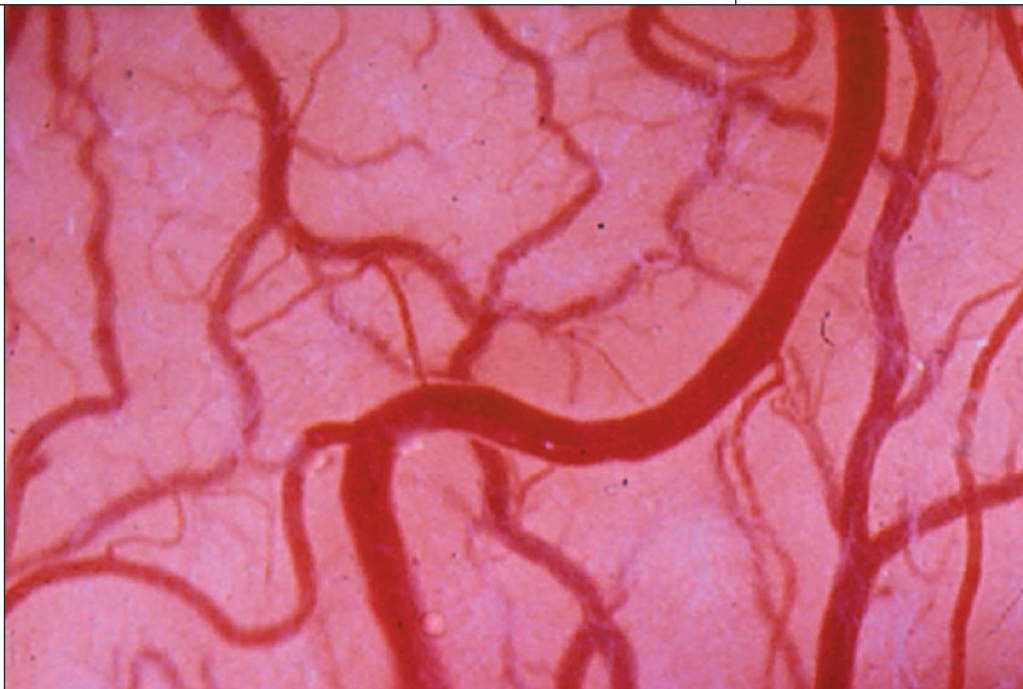
if $c_0 \geq MIC$, where $t_1 = \frac{1}{k} \ln\left(\frac{c_0}{MIC}\right)$ and $t_2 = \frac{\ln 2}{r_2}$.

Applications of Derivatives

4

In the blood vascular system, blood vessels divide into smaller vessels at certain angles and these angles affect the resistance of the blood and therefore the energy expended by the heart in pumping blood. In Example 6 in Section 4.4 we determine the vascular branching angle that minimizes this energy.

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4.1 Maximum and Minimum Values

PROJECT: The Calculus of Rainbows

4.2 How Derivatives Affect the Shape of a Graph

4.3 L'Hospital's Rule: Comparing Rates of Growth

PROJECT: Mutation-Selection Balance in Genetic Diseases

4.4 Optimization Problems

PROJECT: Flapping and Gliding

PROJECT: The Tragedy of the Commons: An Introduction to Game Theory

4.5 Recursions: Equilibria and Stability

4.6 Antiderivatives